eMethods

Echocardiography Data Extraction

As per our previously published echocardiography processing pipeline,[1] pixel data from 2D echocardiographic videos was extracted from the raw Digital Imaging and Communications in Medicine (DICOM) files and deidentified by masking out peripheral pixels containing protected health information. All videos were processed by a pretrained view classifier[2] to determine both the echocardiographic view and imaging mode by randomly selecting ten frames and averaging predicted view probabilities over the ten frames. While the view classifier could discriminate 23 fine-grained view variations, we considered the following key views for later view relevance analysis: apical 2-, 3-, 4-, and 5-chamber (A2C, A3C, etc.), parasternal long axis (PLAX), parasternal short axis (PSAX), right ventricle (RV) inflow, subcostal, and suprasternal. These steps were applied to all YNHHS, RVENet+, and POCUS cohorts.

Automated Color Doppler Detection

To detect the use of color Doppler imaging in the YNHHS cohort, we performed a three-step process of identifying videos that were (i) classified as "Other" by the view classifier, (ii) classified as color Doppler by a custom color Doppler detection model, and (iii) contained a nontrivial amount of red pixels.

For step (ii), we developed a dedicated color Doppler detection model on a manually curated dataset of echocardiogram frames derived from studies not present in the YNHHS dataset used for PanEcho development. Specifically, we manually labeled the presence of color Doppler in all videos from five studies and included videos from another five studies that were known to not contain color Doppler as determined by the view classifier. This dataset of 11,240 labeled frames was then randomly split intro training (80%) and validation (20%) sets at the study level in order to ensure sufficient sample size and diversity of labeled examples in the held-out validation set. An ImageNet-pretrained ConvNeXt-T[3] convolutional neural network (CNN) was trained to classify the presence of color Doppler using a batch size of 128, the Adam optimizer[4] with a learning rate of 0.0001, and a weighted binary cross-entropy loss for ten epochs. All frames were downsampled to 256 x 256 resolution, center cropped to 224 x 224, and normalized with ImageNet channel-wise means and standard deviations. The model achieved 100% accuracy on the validation set and was then applied to all videos classified as "Other" by the view classifier. Similar to view classification, ten randomly selected frames from each video were passed to the color Doppler detection model, and predictions were averaged over the ten frames; videos classified as "Other" but not classified as color Doppler at this step were excluded from the cohort.

As a final quality check, for step (iii), the candidate color Doppler videos underwent color detection to assert the presence of the hue of red typically present in color Doppler echocardiography to indicate blood flow toward the transducer. Frames in each video were converted to the HSV color space, and individual pixels were determined to be red if their HSV values fell between (-10, 150, 150) and (10, 255, 255). Videos were deemed to contain a nontrivial number of red pixels if the total fraction of unique pixels that were red at any point in the video exceeded 1%; videos that passed the previous two steps but failed this step were discarded. Beyond filtering out videos that were neither color Doppler nor 2D B-mode, we did not perform any further quality control to encourage robustness to variations in acquisition quality (e.g., low-contrast or off-axis images), ultrasound machine settings, etc. encountered in real-world clinical practice.

To detect color Doppler in the RVENet+ and POCUS cohorts, we only applied steps (i) and (iii) outlined above for simplicity; the intermediate step (ii) above was needed to obtain reliable color Doppler labels for later view relevance analysis, which was only performed in the YNHHS cohort. For the POCUS cohort, all videos with color Doppler imaging were removed, as this is not a standard acquisition in a cardiac-focused point-of-care ultrasound examination.

Cohort Curation

After color Doppler detection, we limited the YNHHS cohort to contain at most four unique studies per patient – randomly selecting four studies to keep for patients examined at least five times – to prevent overrepresentation of specific patients and outcomes. Next, the resulting dataset was split into development and internal validation sets, with studies performed from July to December 2022 set aside as a temporally distinct test set. The remaining studies from January 2016 to June 2022 were to be used for model development after removing studies from all patients present in the test set to prevent data leakage. The development set was randomly partitioned into training (92.5%) and validation (7.5%) sets at the patient level for model training.

Since the POCUS cohort consisted of emergency department acquisitions by non-experts, greater quality control was needed to ensure minimal diagnostic quality. First, non-cardiac videos from linear array transducers

were detected and removed by identifying videos with significant non-background content in the upper corners. Second, in addition to removing color Doppler acquisitions as described above, all subcostal and suprasternal views were removed as these often did not capture the heart. Finally, only studies with at least three of the following five views represented in the study were retained: PLAX, PSAX at the level of the papillary muscle, A4C, A2C, and A5C. This was done to ensure the inclusion of studies following a basic limited acquisition protocol with key views necessary for the multi-task analysis performed by PanEcho.

Echocardiography Data Preprocessing

In all cohorts, all videos underwent thorough deidentification by masking out pixels beyond the central image content. Namely, each frame was binarized with a fixed threshold, then all pixels outside the convex hull of the largest contour were masked out. All videos were then cropped to the central image content in a temporally consistent manner and downsampled to 256 x 256 resolution with bicubic interpolation. These steps were not necessary for the EchoNet-Dynamic and EchoNet-LVH datasets, which were already deidentified; EchoNet-LVH videos were downsampled to 256 x 256 resolution with bicubic interpolation, and EchoNet-Dynamic videos – already downsampled to 112 x 112 resolution – were upsampled to 256 x 256 resolution with bilinear interpolation to maintain consistent input spatial resolution.

Echocardiographic Reporting Labels

The labels used in this study were determined by reviewing all standard structured echocardiography fields included in the data reporting software (Lumedx®, Oakland, CA) used by YNHHS hospitals. Our clinical team compared these fields against elements from a standard ASE report and identified clinically relevant labels that were consistently captured in YNHHS practice and would provide comprehensive coverage of myocardial size, function, thickness spanning all chambers, as well as valve function and structure and miscellaneous features commonly reported in TTE examinations (e.g., pericardial effusion, great vessels, etc.). We excluded labels where missingness was so extreme that sample sizes would not be sufficient for reliable evaluation (<30 total positive examples for classification tasks) or where missingness was decidedly not-at-random. For instance, we excluded aortic regurgitation pressure half-time, which was generally only reported when considered relevant to patient care.

For YNHHS and POCUS studies, labels were extracted from the local electronic echocardiography reporting system (Lumedx®, Oakland, CA). For the RVENet+ dataset, labels were extracted from the local picture archiving and communication system (Philips Ultrasound Workspace, Philips Medical Systems, Best, the Netherlands) of the Echocardiography Research Laboratory at the Heart and Vascular Center of Semmelweis University; LV volumes and EF were measured using 3D echocardiography (LV-Analysis 3, TomTec Imaging Systems, Unterschleissheim, Germany). For all cohorts, labels reflect the final measurements and reporting confirmed by certified echocardiographers. See **eTable 1** for a description of the guidelines used to determine each categorical diagnostic label; while these guidelines reflect standard practice, the final reporting is ultimately subject to the discretion of the original interpreting cardiologist given the broader clinical context of the patient.

To minimize the effect of extreme outliers on regression tasks, we applied winsorization to all continuous variables, limiting the lowest and highest values to the 0.5 and 99.5 percentile values, respectively. These cutoffs were determined based on the ground truth measurements present in the YNHHS training set and applied to all other cohorts accordingly. Additionally, given the relatively low prevalence of severe phenotypes across certain categorical labels in classification tasks, we pooled moderate and severe phenotypes into shared severity groups for selected tasks. This was done to ensure sufficient sample sizes for reliable model development and evaluation on these rare classes. In the YNHHS and POCUS validation datasets, erroneous LV ejection fraction values were removed (deemed missing) by identifying values entered as 27.5% but having LV Systolic Dysfunction entered as "Mild" or "None".

For the EchoNet-Dynamic and EchoNet-LVH cohorts, a total of 10 labels were extracted based on the measurements provided in these publicly available datasets. EchoNet-Dynamic provided LV ejection fraction, LV ED Volume, and LV ES Volume measurements for each video (one provided per study). Based on this, we determined categorical LV Systolic Dysfunction labels as follows: "None" = LVEF ≥ 54%, "≥ Moderate" = LVEF ≤ 40%, and "Mild" otherwise. EchoNet-LVH provided LVIDd, LVIDs, LVPWd, and IVSd measurements. From these, we determined categorical Increased LV Size labels via ""≥ Moderate" = LVIDd ≥ 6.4 cm, "Normal" = LVIDd ≤ 5.2 cm, and "Mild" otherwise, as well as Increased LV Wall Thickness via ""≥ Moderate" = IVSd ≥ 1.3 cm & LVPWd ≥ 1.3 cm, "Any" = IVSd ≥ 1.1 cm & LVPWd ≥ 1.1 cm, and "None" otherwise. While standard cutoffs for these conditions are sex-dependent, these conservative thresholds were chosen since patient sex was not provided in EchoNet-LVH nor EchoNet-Dynamic.

PanEcho Implementation Details

In designing the architecture for PanEcho, we adopted a decoupled "2+1D" approach to modeling echocardiogram videos – with separate modules to learn spatial and temporal features – primarily for downstream flexibility; for instance, our 2D image backbone can be readily adapted for any echocardiographic task, while a 3D backbone would be more difficult to retrofit to a 2D image-only task such as segmentation. Each video frame is first processed by the 2D image encoder, an ImageNet[5]-pretrained ConvNeXt-T[3] convolutional neural network (CNN), which produces a learned feature vector, or representation, of each frame. These frame-wise representations are then interpreted as an ordered sequence – like words in a sentence in natural language processing – and modeled using self-attention[6] to learn time-varying associations over the frames. Frame order is embedded via sinusoidal positional encoding, which is then elementwise added to the frame-wise feature vectors and fed to a Transformer encoder consisting of four layers, each with eight self-attention heads. Mean pooling is then used to aggregate frame-wise feature vectors into a single video-level representation, which is used as input to the task-specific output heads. Each output head consists of a Dropout[7] layer with probability 0.25 and a fully-connected layer. Both regression and binary classification tasks used one output neuron, the latter followed by a sigmoid activation. Multi-class classification tasks with k classes used k output neurons with softmax activation, and multi-label classification tasks were modeled with separate binary classification heads for each class. PanEcho was trained to minimize the mean of all valid task-specific losses – cross-entropy for classification tasks and mean squared error for regression tasks. Studies with missing labels were masked from task-specific loss computation, and tasks with fully missing labels (i.e., no studies in the mini-batch had ground truth labels) were masked from aggregated loss computation. To control for varying units and scales of regression tasks, we first divided each regression loss by the mean observed value of that measurement in the training set before loss aggregation.

PanEcho was implemented and trained in PyTorch[8] version 2.1.0 with distributed training across eight NVIDIA A100 graphics processing units (GPUs) with automatic mixed precision to maximize throughput. During training, the model received as input a randomly sampled video clip of 16 consecutive frames from an echocardiogram, following prior work.[1] To increase robustness to variations in acquisition and increase effective sample size, the following augmentations were performed to all video frames in a temporally consistent manner: random crop to 224 x 224 resolution, random horizontal flip with probability 0.5, random rotation within (-15°, 15°), then followed by ImageNet normalization. The model was trained with a batch size of 16 per GPU, the Adam optimizer[4], and minimized the multi-task loss described above with learning rate 0.0001. The learning rate was reduced by a factor of 0.5 if the validation metric (mean classification AUC and regression R^2 across all tasks) did not improve for three consecutive epochs; though MAE was the primary evaluation metric for regression tasks, this validation metric was chosen because AUC and R^2 are both increasing and bounded to $[0, 1]$. The model was trained for a maximum of 30 epochs with early stopping if validation metric did not improve for 10 consecutive epochs. At test time, four 16-frame clips are randomly sampled from each video and task-wise predictions are averaged over all clips to produce video-level predictions. Since PanEcho is view-agnostic and labels are determined at the study level, predictions from all videos acquired during the same study (regardless of imaging mode or view) were averaged to form a single study-level prediction for each task.

Validation on Diagnostic Echocardiography

PanEcho was evaluated on all available labels in the internal YNHHS validation set (39) and the external RVENet+ (38), EchoNet-Dynamic (4), and EchoNet-LVH (6) cohorts. While all 39 labels were extracted for the RVENet+ cohort, there were no instances of Elevated LVOT PG in the cohort, meaning performance metrics could not be computed for this task. As the YNHHS and RVENet+ were multi-view datasets including all available 2D echocardiographic views, video-level predictions for all acquisitions in a study were averaged to produce holistic study-level predictions for each task at test time. EchoNet-Dynamic and EchoNet-LVH only provided individual videos – from the A4C and PLAX views, respectively – meaning PanEcho performance was determined based on one video per study.

Validation on Point-of-care Echocardiography

PanEcho was additionally evaluated on a simulated abbreviated TTE dataset and a real-world POCUS dataset from bedside acquisitions by non-experts. The Abbreviated TTE dataset was derived from the internal YNHHS validation set by removing all color Doppler videos and selecting at most one videos from the following five key views per study: PLAX, PSAX at the level of the papillary muscle, A4C, A5C, and A2C. While the view classifier was capable of discriminating variations of several view types (e.g., A4C vs. A4C – LV occluded vs. A4C – LA occluded), only videos from the "primary" view were considered for this analysis. In the case that multiple videos

from one of the five key views were present in a study, the video with the highest predicted view probability was selected.

Task-specific View Relevance Analysis

We defined the normalized view relevance score

$$R_{v,t} = \begin{cases} \frac{m_{v,t}}{\max_v (m_{v,t})}, & t \text{ is a classification task} \\ \frac{m_{v,t}}{\min_v (m_{v,t})}, & t \text{ is a regression task} \end{cases},$$

where $R_{v,t}$ is the relevance of view v for task t, and $m_{v,t}$ is the performance metric on task t when only using view v (AUC for classification tasks and MAE for regression tasks). This produces a task-normalized score where, for a given task, 1 represents the most informative view, and each score can be interpreted as the "fractional importance relative to the best view." This analysis was performed on the internal YNHHS validation set and metrics were computed after selecting a maximum of three videos per view in a given study with the most confident predicted view probability by the view classifier; this was done to control for the variable prevalence of views – without this, the most common views would be overrepresented within each study and unfairly benefit from a greater ensembling effect after video-level aggregation. For tasks typically performed with or aided by some form of Doppler imaging, we performed this analysis again after including color Doppler videos (from any echocardiographic view) as an additional "view" to assess the task-dependent value of color Doppler imaging. Detailed task-specific view relevance scores are visualized in **eFigure 4**.

Figure 5 summarizes view relevance for aspects of cardiovascular health by averaging view relevance scores across tasks falling under each category. Specifically, we include the following categories (constituent tasks in parentheses): *Left Ventricular Structure* (IVSd, LVPWd, LVIDs, LVIDd, Increased LV Size [≥ Moderate], Increased LV Wall Thickness [≥ Moderate], LV ED Volume, LV ES Volume), *Left Ventricular Function* (LV Ejection Fraction, LV Stroke Volume, LV Systolic Dysfunction [≥ Moderate], LV Diastolic Dysfunction [≥ Moderate], LV Wall Motion Abnormalities), *Right Ventricle* (RVIDd, TAPSE, RV Systolic Pressure, Increased RV Size [≥ Moderate], RV Systolic Dysfunction), *Left Atrium* (LAIDs, LA Volume, Increased LA Size), *Aortic Valve* (Aortic Root Dimension, AV Structure, AV Stenosis [Severe], AV Regurgitation [≥ Moderate]), *Valvular Regurgitation* (AV Regurgitation [≥ Moderate], MV Regurgitation [≥ Moderate], TV Regurgitation [≥ Moderate]).

Group Fairness Analysis

Beyond predictive performance, we also evaluate the demographic parity of PanEcho across protected subgroups of sex and race to assess bias (**eFigure 3**). This analysis involved computation of the difference in AUC (ΔAUC) between subgroups for each classification task in the internal YNHHS validation set. Groupwise comparisons were excluded if the subgroup has <30 positive examples of a particular label to ensure sufficient sample size for reliable evaluation. For comparisons across sex, this meant that analysis could be performed for 15/18 labels; for comparisons across race, analysis could only be performed for 12/18 labels for the White or Caucasian, Black or African American, and Other racial groups. If the 95% CI for ΔAUC overlapped with zero, then PanEcho was considered to demonstrate unbiased predictive performance between those particular subgroups on a given task.

Image Quality Analysis

To evaluate PanEcho's robustness to variations in image quality, we assessed its LV ejection fraction estimation performance across images of different quality levels in the internal YNHHS validation set (**eFigure 4**). While there is no standard metric for image quality, we consider the view classifier's predicted view probability – interpreted as its "confidence" – as a proxy for image quality. For this analysis, only videos from the A4C view were included, as this is the key view for LV ejection fraction evaluation. Similar to the view relevance analysis, a maximum of three A4C videos per study were included (selected by maximum view classifier confidence) per study to control for the number of videos used to make study-level predictions. Based on the mean A4C view classifier probability of the ≤3 videos included in each study, five equal-sized bins (quintiles) of image quality were identified: Q1 (0.162, 0.667], Q2 (0.667, 0.847], Q3 (0.847, 0.965], Q4 (0.965, 0.998], and Q5 (0.998, 1]. LV ejection fraction estimation was then evaluated for studies falling within each quintile by MAE.

eTable 1 | Diagnostic Label Definitions and Guidelines

	Class	Suggested Definition
AV Regurgitation	*None, Mild, ≥ Moderate*	Per recommendations[9]
AV Stenosis	*None, Mild-Moderate, Severe*	Per recommendations[10,11]
AV Structure	*Bicuspid*	-
Increased LV Size	*None*	Female: LV EDVi ≤61 mL/m^2, LVIDd ≤5.2 cm Male: LV EDVi ≤74 mL/m^2, LVIDd ≤5.8 cm
	Mild	Female: LV EDVi 62-70 mL/m^2, LVIDd 5.3-5.6 cm Male: LV EDVi 75-89 mL/m^2, LVIDd 5.9-6.3 cm
	≥ Moderate	Female: LV EDVi ≥71 mL/m^2, LVIDd ≥5.7 cm Male: LV EDVi ≥90 mL/m^2, LVIDd ≥6.4 cm
Increased LV Wall Thickness	*Any*	Female: IVSd & LVPWd ≥1.0 cm, alternatively LVMi >95 g/m^2 Male: IVSd & LVPWd ≥1.1 cm, or LVMi >115 g/m^2
	≥ Moderate	Female: IVSd & LVPWd ≥1.2 cm; Male: IVSd & LVPWd ≥1.3 cm
LV Systolic Dysfunction	*None*	Female: LV EF ≥54%; Male: LV EF ≥ 52%
	Mild	Female: LV EF 41-53%; Male: LV EF 41-51%
	≥ Moderate	Female & Male: LV EF ≤40%
LV Diastolic Dysfunction	*None*	Per recommendations[12]
	Mild (~ grade 1)	
	≥ Moderate (~ grade 2, 3)	
LV Wall Motion Abnormalities	*Any*	-
Increased LA Size	*None*	LAVi≤34 mL/m^2
	Mild	LAVi 35-41 mL/m^2
	≥ Moderate	LAVi ≥42 mL/m^2
Elevated LVOT PG	*≥ 20 mmHg*	-
MV Stenosis	*Any (≥ Mild)*	Per guidelines[11]
MV Regurgitation	*None, Mild, ≥ Moderate*	Per guidelines[9]
Pericardial Effusion	*Any (≥ Mild)*	Excluding trivial effusions, this includes an effusion which is <10 mm but can be seen throughout diastole and systole (moderate 10-20 mm, and large >20 mm)
Elevated RA Pressure	*≥ 8 mmHg*	E.g., dilated inferior vena cava >2.1 cm with preserved collapsibility >50% with sniff
Increased RA Size	*Any*	Minor axis length ≥4.6 cm
Increased RV Size	*None*	RV diameter at mid-ventricle ≤3.5 cm
	Mild	RV diameter at mid-ventricle 3.6-3.8 cm
	≥ Moderate	RV diameter at mid-ventricle ≥3.9 cm. For RV EDVi consider ≥160 mL/m^2 or RV ESVi ≥ 80 mL/m^2 to define moderate or severe dilation.
RV Systolic Dysfunction	*Any*	RV EF ≤ 40%
TV Regurgitation	*None, Mild, ≥ Moderate*	Per recommendations[9]

AV = aortic valve; EDVi = end-diastolic volume index; EF = ejection fraction; ESVi = end-systolic volume index; IVSd = interventricular septal thickness in diastole; LA = left atrium; LAVi = left atrial volume index; LV = left ventricle; LVIDd = left ventricular internal diameter at diastole; LVIDs = left ventricular internal diameter at systole; LVMi = left ventricular mass index; LVOT = left ventricular outflow tract; LVPWd = left ventricular posterior wall thickness in diastole; MV = mitral valve; PG = pressure gradient; RA = right atrium; RV = right ventricle; TV = tricuspid regurgitation.

223 **eTable 2 | Detailed Description of Study Cohorts**

		YNHHS				RVENet+	POCUS
		Training	**Tuning**	**Validation**	**Abbreviated TTE**		
Dates Sampled		1/2016-6/2022	1/2016-6/2022	7/2022-12/2022	7/2022-12/2022	11/2013-3/2021	9/2015-11/2023
Patients	***Total**, n*	18,343	1,474	4,588	3,107	831	3,170
Studies	***Total**, n*	25,130	2,005	5,130	3,608	944	3,310
Videos	***Total**, n*	999,727	79,869	114,280	9,091	18,862	25,407
	***Per Study**, median (IQR)*	39.0 (30.0-48.0)	39.0 (31.0-48.0)	17.0 (12.0-31.0)	2.0 (1.0-4.0)	19.5 (17.0-22.0)	6.0 (5.0-9.0)
Age (years)	***Present**, median (IQR)*	69.0 (58.0-79.0)	68.0 (58.0-79.0)	68.0 (57.0-78.0)	68.0 (57.0-78.0)	50.0 (25.0-66.0)	66.0 (53.0-79.0)
	***Missing**, n*	0	0	0	0	0	0
Sex	***Male**, n (%)*	13,184 (52.5)	1,050 (52.4)	2,585 (50.4)	1,846 (51.2)	586 (62.1)	1,817 (54.9)
	***Female**, n (%)*	11,946 (47.5)	955 (47.6)	2,545 (49.6)	1,762 (48.8)	358 (37.9)	1,493 (45.1)
	***Missing**, n*	0	0	0	0	0	0
Race	***White**, n (%)*	19,333 (80.8)	1,541 (81.4)	3,893 (80.6)	2,728 (80.0)	944 (100)	2,073 (65.8)
	***Black**, n (%)*	3,455 (14.4)	256 (13.5)	679 (14.0)	501 (14.7)	0 (0)	871 (27.7)
	***Asian**, n (%)*	403 (1.7)	33 (1.7)	112 (2.3)	79 (2.3)	0 (0)	68 (2.2)
	***American Indian**, n (%)*	60 (0.3)	8 (0.4)	15 (0.3)	11 (0.3)	0 (0)	12 (0.4)
	***Pacific Islander**, n (%)*	639 (2.7)	54 (2.9)	124 (2.6)	89 (2.6)	0 (0)	115 (3.7)
	***Other**, n (%)*	36 (0.2)	0 (0)	10 (0.2)	3 (0.1)	0 (0)	10 (0.3)
	***Missing**, n*	1204	113	297	197	0	161
Ethnicity	***Hispanic**, n (%)*	1,947 (8.0)	147 (7.5)	472 (9.5)	313 (9.0)	0 (0)	372 (11.4)
	***Non-Hispanic**, n (%)*	22,511 (92.0)	1,805 (92.5)	4,486 (90.5)	3,178 (91.0)	944 (100)	2,884 (88.6)
	***Missing**, n*	672	53	172	117	0	54
Patient Status	***Outpatient**, n (%)*	15,569 (62.4)	1,273 (64.0)	3,511 (68.7)	2,584 (71.9)	524 (55.5)	0 (0)
	***Inpatient**, n (%)*	7,807 (31.3)	591 (29.7)	1,296 (25.4)	813 (22.6)	420 (44.5)	0 (0)
	***Observation**, n (%)*	1,519 (6.1)	119 (6.0)	291 (5.7)	189 (5.3)	0 (0)	0 (0)
	***Emergency**, n (%)*	58 (0.2)	5 (0.3)	11 (0.2)	7 (0.2)	0 (0)	3,310 (100)
	***Missing**, n*	177	17	21	15	0	0
Hypertension	***Yes**, n (%)*	18,535 (73.8)	1,447 (72.2)	3,577 (69.7)	2,552 (70.7)	372 (39.4)	2,323 (70.2)
	***Missing**, n*	0	0	0	0	0	0
Diabetes	***Yes**, n (%)*	6,774 (27.0)	492 (24.5)	1,243 (24.2)	866 (24.0)	119 (12.6)	1,001 (30.2)
	***Missing**, n*	0	0	0	0	0	0
Dyslipidemia	***Yes**, n (%)*	5,367 (21.4)	387 (19.3)	1,164 (22.7)	865 (24.0)	239 (25.3)	493 (14.9)
	***Missing**, n*	0	0	0	0	0	0
Chronic Kidney Disease	***Yes**, n (%)*	2,394 (9.5)	182 (9.1)	350 (6.8)	251 (7.0)	108 (11.4)	442 (13.4)
	***Missing**, n*	0	0	0	0	0	0
Coronary Artery Disease	***Yes**, n (%)*	5,125 (20.4)	395 (19.7)	824 (16.1)	578 (16.0)	91 (9.6)	1,103 (33.3)
	***Missing**, n*	0	0	0	0	0	0

Aortic Root Diameter (mm)	***Present**, median (IQR)*	33.0 (30.0-36.0)	33.0 (30.0-36.3)	33.0 (30.0-36.6)	33.0 (30.0-37.0)	32.0 (30.0-34.2)	33.0 (30.0-36.0)
	***Missing**, n*	2,041	165	162	128	1	872
AV Peak Velocity (m/s)	***Present**, median (IQR)*	1.5 (1.3-2.1)	1.5 (1.2-2.0)	1.4 (1.2-1.8)	1.5 (1.2-1.9)	1.3 (1.2-1.6)	1.4 (1.2-1.7)
	***Missing**, n*	2,644	205	631	367	10	876
AV Regurgitation	***None-Trace**, n (%)*	16,975 (75.5)	1,363 (75.3)	3,676 (77.9)	2,491 (75.5)	773 (81.9)	2,042 (78.4)
	***Mild**, n (%)*	3,980 (17.7)	329 (18.2)	773 (16.4)	591 (17.9)	140 (14.8)	398 (15.3)
	***≥ Moderate**, n (%)*	1,526 (6.8)	119 (6.6)	272 (5.8)	218 (6.6)	31 (3.3)	164 (6.3)
	***Missing**, n*	2,649	194	409	308	0	706
AV Stenosis	***None**, n (%)*	17,118 (79.5)	1,422 (81.8)	4,214 (89.6)	2,907 (88.5)	850 (90.0)	2,109 (91.3)
	***Mild-Moderate**, n (%)*	2,616 (12.1)	192 (11.0)	401 (8.5)	309 (9.4)	9 (1.0)	164 (7.1)
	***Severe**, n (%)*	1,809 (8.4)	125 (7.2)	86 (1.8)	69 (2.1)	85 (9.0)	38 (1.6)
	***Missing**, n*	3,587	266	429	323	0	999
AV Structure	***Bicuspid**, n (%)*	207 (1.0)	5 (0.3)	52 (1.1)	37 (1.1)	8 (0.8)	10 (0.4)
	***Missing**, n*	4,053	296	571	389	0	743
LV Ejection Fraction (%)	***Present**, median (IQR)*	62.0 (56.1-66.9)	62.5 (57.0-66.7)	61.4 (57.0-65.0)	61.4 (57.0-65.2)	57.5 (52.2-61.4)	57.0 (40.0-63.0)
	***Missing**, n*	942	69	85	58	0	116
E|e' Ratio	***Present**, median (IQR)*	10.4 (7.9-14.3)	9.9 (7.8-13.7)	9.2 (7.2-12.1)	9.3 (7.2-12.3)	7.0 (5.5-10.6)	10.6 (7.9-15.0)
	***Missing**, n*	6,085	462	781	599	216	1,455
Global Longitudinal Strain (%)	***Present**, median (IQR)*	18.0 (14.0-20.0)*	18.0 (15.0-20.0)*	19.0 (18.0-21.0)*	19.0 (17.0-21.0)*	17.6 (13.3-20.7)*	16.0 (11.0-19.0)*
	***Missing**, n*	23,663	1,888	4,669	3,297	78	3,137
IVSd (mm)	***Present**, median (IQR)*	10.9 (9.1-12.6)	10.8 (9.0-12.5)	10.0 (8.8-11.5)	10.0 (8.9-11.7)	11.0 (9.0-12.0)	10.0 (8.9-11.7)
	***Missing**, n*	1,178	82	85	64	0	611
LAIDs (mm)	***Present**, median (IQR)*	40.0 (35.0-46.0)	40.0 (34.7-45.0)	39.6 (35.0-45.0)	40.0 (35.0-45.5)	40.0 (34.0-46.0)	41.0 (36.0-48.0)
	***Missing**, n*	2,861	244	298	220	343	940
Increased LA Size	***None**, n (%)*	15,595 (62.1)	1,251 (62.4)	3,527 (68.8)	2,440 (67.6)	447 (52.9)	2,166 (65.4)
	***Mild**, n (%)*	4,543 (18.1)	354 (17.7)	805 (15.7)	570 (15.8)	125 (14.8)	403 (12.2)
	***≥ Moderate**, n (%)*	4,992 (19.9)	400 (20.0)	798 (15.6)	598 (16.6)	273 (32.3)	741 (22.4)
	***Missing**, n*	0	0	0	0	99	0
LA Volume (cm^3)	***Present**, median (IQR)*	58.5 (45.0-74.4)	58.5 (45.0-74.0)	55.2 (43.0-70.0)	56.0 (43.9-71.0)	64.5 (46.2-91.3)	61.9 (47.0-79.0)
	***Missing**, n*	7,143	554	1,023	730	99	1,446
LV Diastolic Dysfunction	***None**, n (%)*	8,583 (48.4)	702 (50.1)	2,524 (62.0)	1,686 (59.7)	485 (58.7)	880 (49.7)
	***Mild**, n (%)*	6,376 (36.0)	496 (35.4)	1,214 (29.8)	892 (31.6)	182 (22.0)	473 (26.7)
	***≥ Moderate**, n (%)*	2,762 (15.6)	203 (14.5)	333 (8.2)	247 (8.7)	159 (19.2)	416 (23.5)
	***Missing**, n*	7,409	604	1,059	783	118	1,541
LV ED Volume (cm^3)	***Present**, median (IQR)*	96.9 (72.0-130.0)	97.4 (72.0-127.0)	105.1 (80.6-136.9)	105.0 (80.8-137.0)	128.9 (104.0-166.1)	130.0 (98.0-170.0)
	***Missing**, n*	6,053	505	1,681	1,084	0	821
LV ES Volume (cm^3)	***Present**, median (IQR)*	36.0 (25.2-53.3)	36.4 (25.0-51.0)	40.1 (28.9-56.4)	40.0 (28.9-56.4)	55.0 (41.9-78.0)	57.0 (38.0-92.4)
	***Missing**, n*	6,243	515	1,695	1,096	0	858

LVIDd (mm)	***Present**, median (IQR)*	45.5 (41.0-50.1)	45.1 (40.5-50.0)	46.0 (41.8-50.7)	46.0 (41.8-50.7)	49.0 (45.0-55.0)	47.2 (42.1-52.8)
	***Missing**, n*	1,170	85	91	69	0	614
LVIDs (mm)	***Present**, median (IQR)*	29.5 (25.7-34.1)	29.4 (25.6-33.7)	30.0 (26.2-34.0)	30.0 (26.3-34.0)	33.0 (28.0-38.0)	32.2 (27.6-39.0)
	***Missing**, n*	1,323	94	102	74	130	636
Elevated LVOT PG	***≥ 20 mmHg**, n (%)*	249 (1.2)	25 (1.5)	24 (0.6)	22 (0.7)	0 (0)	2 (0.1)
	***Missing**, n*	4,600	374	776	512	0	877
LVOT Diameter (mm)	***Present**, median (IQR)*	20.0 (19.0-21.8)	20.0 (19.0-22.0)	20.0 (19.0-22.0)	20.0 (19.0-22.0)	20.2 (18.9-22.0)	20.0 (19.0-22.0)
	***Missing**, n*	12,878	1,046	1,411	1,037	330	2,019
LVPWd (mm)	***Present**, median (IQR)*	10.4 (9.0-12.0)	10.4 (9.0-12.0)	10.0 (8.7-11.2)	10.0 (8.8-11.3)	9.0 (8.0-10.0)	10.0 (8.9-11.4)
	***Missing**, n*	1,189	85	90	68	0	612
Increased LV Size	***None**, n (%)*	16,249 (86.2)	1,329 (88.4)	3,660 (85.8)	2,527 (84.8)	774 (82.0)	1,734 (65.9)
	***Mild**, n (%)*	1,396 (7.4)	93 (6.2)	338 (7.9)	253 (8.5)	81 (8.6)	351 (13.3)
	***≥ Moderate**, n (%)*	1,201 (6.4)	82 (5.5)	270 (6.3)	201 (6.7)	89 (9.4)	547 (20.8)
	***Missing**, n*	6,284	501	862	627	0	678
LV Stroke Volume (cm^3)	***Present**, median (IQR)*	57.0 (43.0-74.0)	57.5 (43.2-75.4)	62.0 (48.3-79.0)	62.0 (48.1-79.0)	69.5 (56.4-84.6)	64.0 (49.1-80.5)
	***Missing**, n*	6,129	516	1,735	1,126	0	833
LV Systolic Dysfunction	***None**, n (%)*	19,820 (84.5)	1,607 (85.9)	4,275 (87.6)	3,005 (87.4)	692 (73.3)	1,929 (61.9)
	***Mild**, n (%)*	1,782 (7.6)	124 (6.6)	319 (6.5)	223 (6.5)	130 (13.8)	430 (13.8)
	***≥ Moderate**, n (%)*	1,849 (7.9)	140 (7.5)	284 (5.8)	212 (6.2)	122 (12.9)	755 (24.2)
	***Missing**, n*	1,679	134	252	168	0	196
LV Wall Motion Abnormalities	***Any**, n (%)*	1,704 (20.6)	143 (21.4)	348 (22.6)	249 (22.7)	116 (12.3)	630 (44.5)
	***Missing**, n*	16,855	1,338	3,589	2,512	0	1,895
Increased LV Wall Thickness	***Any**, n (%)*	12,509 (54.8)	990 (54.0)	1,971 (40.2)	1,465 (42.7)	232 (24.6)	1,206 (44.1)
	***≥ Moderate**, n (%)*	3,021 (13.2)	239 (13.0)	315 (6.4)	250 (7.3)	46 (4.9)	258 (9.4)
	***Missing**, n*	2,324	172	229	174	0	575
MV Regurgitation	***None-Trace**, n (%)*	12,041 (50.7)	996 (52.3)	2,976 (59.4)	1,967 (56.0)	479 (57.8)	1,441 (51.5)
	***Mild**, n (%)*	7,968 (33.6)	626 (32.9)	1,419 (28.3)	1,062 (30.2)	209 (25.2)	815 (29.1)
	***≥ Moderate**, n (%)*	3,718 (15.7)	282 (14.8)	612 (12.2)	482 (13.7)	141 (17.0)	544 (19.4)
	***Missing**, n*	1,403	101	123	97	115	510
MV Stenosis	***Any**, n (%)*	758 (4.1)	54 (3.7)	89 (2.1)	67 (2.3)	11 (1.3)	71 (3.4)
	***Missing**, n*	6,823	542	928	693	115	1,248
Pericardial Effusion	***Any**, n (%)*	935 (4.0)	71 (3.8)	21 (3.4)	21 (3.4)	47 (5.0)	305 (10.9)
	***Missing**, n*	1,608	121	4,507	2,989	0	514
RA Transverse Dimension (mm)	***Present**, median (IQR)*	37.2 (33.0-43.0)	38.0 (33.0-43.0)	37.0 (33.0-42.0)	38.0 (33.0-43.0)	40.5 (35.0-45.0)	40.0 (34.0-46.0)
	***Missing**, n*	14,163	1,147	2,233	1,541	0	1,749
Elevated RA Pressure	***≥ 8 mmHg**, n (%)*	4,596 (21.0)	344 (19.5)	722 (15.5)	485 (14.7)	531 (56.2)	1,200 (43.2)
	***Missing**, n*	3,196	239	484	319	0	535
Increased RA Size	***Any**, n (%)*	2,988 (29.8)	231 (28.2)	507 (29.3)	374 (29.5)	220 (23.3)	541 (41.3)

	***Missing**, n*	15,115	1,186	3,398	2,341	0	2,000
RVIDd (mm)	***Present**, median (IQR)*	31.0 (27.1-35.1)	31.0 (27.2-35.0)	30.9 (27.0-34.9)	31.0 (27.0-35.1)	31.0 (28.0-35.0)	32.0 (28.2-36.3)
	***Missing**, n*	9,632	791	1,700	1,196	0	999
Increased RV Size	***None**, n (%)*	19,229 (83.8)	1,551 (85.3)	4,088 (85.9)	2,843 (84.6)	738 (78.2)	1,955 (70.0)
	***Mild**, n (%)*	2,313 (10.1)	170 (9.3)	457 (9.6)	359 (10.7)	94 (10.0)	427 (15.3)
	***≥ Moderate**, n (%)*	1,417 (6.2)	98 (5.4)	215 (4.5)	158 (4.7)	112 (11.9)	411 (14.7)
	***Missing**, n*	2,171	186	370	248	0	517
RV Systolic Pressure (mmHg)	***Present**, median (IQR)*	29.0 (23.0-38.0)	28.0 (22.4-37.0)	27.5 (22.5-35.0)	27.1 (22.1-34.4)	28.0 (22.0-38.0)	37.0 (28.0-49.0)
	***Missing**, n*	7,923	589	1,832	1,192	255	1,373
RV S' (cm/s)	***Present**, median (IQR)*	12.3 (10.4-14.4)	12.5 (10.9-14.6)	12.6 (10.9-14.7)	12.6 (10.9-14.6)	13.0 (11.0-15.0)	11.9 (9.5-14.0)
	***Missing**, n*	10,403	864	1,499	1,075	113	1,349
RV Systolic Dysfunction	***Any**, n (%)*	2,589 (11.3)	196 (10.8)	371 (7.8)	263 (7.9)	76 (8.1)	819 (29.1)
	***Missing**, n*	2,243	198	387	261	0	497
TAPSE (mm)	***Present**, median (IQR)*	21.0 (17.9-24.9)	21.5 (18.0-25.0)	21.2 (18.0-24.8)	21.1 (18.0-24.8)	21.0 (17.0-26.0)	19.0 (16.0-23.0)
	***Missing**, n*	9,774	798	1,396	971	0	1,279
TV Peak Gradient (mmHg)	***Present**, median (IQR)*	25.0 (19.6-32.7)	24.0 (19.0-31.4)	24.0 (19.0-30.0)	23.9 (19.0-30.0)	22.0 (17.0-30.0)	29.0 (22.0-40.0)
	***Missing**, n*	6,493	488	1,615	1,042	255	1,142
TV Regurgitation	***None-Trace**, n (%)*	12,301 (50.9)	1,004 (52.1)	2,886 (56.9)	1,941 (54.4)	532 (56.4)	1,430 (46.7)
	***Mild**, n (%)*	8,199 (33.9)	630 (32.7)	1,580 (31.1)	1,173 (32.9)	310 (32.8)	939 (30.7)
	***≥ Moderate**, n (%)*	3,666 (15.2)	294 (15.2)	610 (12.0)	454 (12.7)	102 (10.8)	692 (22.6)
	***Missing**, n*	964	77	54	40	0	249

Demographics and summary statistics of all cohorts used in this study. All summary statistics are computed after excluding missing values and provided at the study level since age and labels are unique to each echocardiography study. * = Global longitudinal strain values are presented as positive to aid readability.

AV = aortic valve; ED = end-diastolic; ES = end-systolic; IVSd = interventricular septum thickness at diastole; IQR = interquartile range; LA = left atrium; LV = left ventricle; LAIDs = left atrial internal diameter at systole; LVIDd = left ventricular internal diameter at diastole; LVIDs = left ventricular internal diameter at systole; LVOT = left ventricular outflow tract; LVPWd = left ventricular posterior wall thickness at diastole; PG = pressure gradient; RA = right atrium; RV = right ventricle; RVIDd = right ventricular internal diameter at diastole; RV S' = right ventricular systolic excursion velocity; TAPSE = tricuspid annular plane systolic excursion; TTE = transthoracic echocardiography; TV = tricuspid valve; YNHHS = Yale-New Haven Health System.

eTable 3 | Detailed Multi-task Classification Performance Evaluation

		Cohort	AUC	AP	Sensitivity	Specificity	Brier Score	n/N
AV Regurgitation	*None-Trace*	YNHHS	0.86 (0.85, 0.87)	0.95 (0.95, 0.96)	0.76 (0.75, 0.77)	0.81 (0.79, 0.83)	0.12 (0.12, 0.13)	3,676/4,721
		RVENet+	0.83 (0.80, 0.85)	0.95 (0.93, 0.96)	0.72 (0.69, 0.74)	0.79 (0.73, 0.84)	0.12 (0.11, 0.13)	773/944
	Mild	YNHHS	0.79 (0.78, 0.80)	0.40 (0.37, 0.43)	0.84 (0.82, 0.86)	0.60 (0.59, 0.61)	0.12 (0.11, 0.12)	773/4,721
		RVENet+	0.79 (0.75, 0.82)	0.40 (0.34, 0.48)	0.81 (0.76, 0.87)	0.62 (0.59, 0.65)	0.11 (0.10, 0.12)	140/944
	≥ Moderate	YNHHS	0.92 (0.91, 0.93)	0.52 (0.47, 0.57)	0.85 (0.82, 0.89)	0.83 (0.82, 0.84)	0.04 (0.04, 0.05)	272/4,721
		RVENet+	0.85 (0.81, 0.89)	0.15 (0.10, 0.25)	0.77 (0.64, 0.90)	0.76 (0.74, 0.78)	0.03 (0.02, 0.04)	31/944
	Mean	YNHHS	0.86 (0.85, 0.87)	0.62 (0.61, 0.64)	0.82 (0.80, 0.83)	0.75 (0.74, 0.75)	0.10 (0.09, 0.10)	N/A
		RVENet+	0.82 (0.80, 0.85)	0.50 (0.48, 0.54)	0.77 (0.72, 0.82)	0.72 (0.70, 0.74)	0.09 (0.08, 0.09)	N/A
AV Stenosis	*None*	YNHHS	0.95 (0.94, 0.96)	0.99 (0.99, 0.99)	0.91 (0.91, 0.92)	0.85 (0.82, 0.88)	0.05 (0.05, 0.05)	4,214/4,701
		RVENet+	0.98 (0.96, 0.99)	1.00 (0.99, 1.00)	0.88 (0.87, 0.90)	0.95 (0.90, 0.98)	0.03 (0.03, 0.04)	850/944
	Mild-Moderate	YNHHS	0.93 (0.92, 0.94)	0.54 (0.50, 0.59)	0.82 (0.79, 0.85)	0.90 (0.89, 0.90)	0.06 (0.05, 0.06)	401/4,701
		RVENet+	0.70 (0.57, 0.81)	0.02 (0.01, 0.04)	0.44 (0.14, 0.73)	0.80 (0.78, 0.82)	0.03 (0.02, 0.03)	9/944
	Severe	YNHHS	0.98 (0.98, 0.99)	0.58 (0.48, 0.67)	0.94 (0.89, 0.98)	0.95 (0.94, 0.95)	0.01 (0.01, 0.01)	86/4,701
		RVENet+	1.00 (0.99, 1.00)	0.97 (0.96, 0.99)	0.96 (0.93, 0.99)	0.98 (0.97, 0.99)	0.04 (0.04, 0.05)	85/944
	Mean	YNHHS	0.96 (0.95, 0.96)	0.70 (0.67, 0.74)	0.89 (0.87, 0.91)	0.90 (0.89, 0.91)	0.04 (0.04, 0.04)	N/A
		RVENet+	0.89 (0.85, 0.93)	0.66 (0.66, 0.67)	0.76 (0.66, 0.86)	0.91 (0.89, 0.92)	0.03 (0.03, 0.04)	N/A
AV Structure	*Bicuspid*	YNHHS	0.81 (0.75, 0.86)	1.00 (1.00, 1.00)	0.70 (0.69, 0.71)	0.69 (0.58, 0.80)	0.01 (0.01, 0.01)	52/4,559
		RVENet+	0.83 (0.65, 0.96)	0.14 (0.03, 0.39)	0.88 (0.67, 1.00)	0.60 (0.58, 0.63)	0.01 (0.00, 0.01)	8/944
Increased LA Size	*Mild*	YNHHS	0.75 (0.74, 0.77)	0.36 (0.33, 0.38)	0.67 (0.65, 0.70)	0.70 (0.69, 0.72)	0.12 (0.11, 0.12)	805/5,130
		RVENet+	0.65 (0.60, 0.69)	0.23 (0.19, 0.29)	0.52 (0.44, 0.60)	0.64 (0.62, 0.67)	0.13 (0.12, 0.14)	125/845
	≥ Moderate	YNHHS	0.89 (0.88, 0.90)	0.59 (0.56, 0.63)	0.79 (0.76, 0.81)	0.83 (0.82, 0.84)	0.09 (0.09, 0.09)	798/5,130
		RVENet+	0.90 (0.88, 0.92)	0.77 (0.73, 0.83)	0.78 (0.74, 0.82)	0.88 (0.86, 0.91)	0.16 (0.15, 0.17)	273/845
	None	YNHHS	0.84 (0.83, 0.85)	0.92 (0.91, 0.92)	0.74 (0.73, 0.75)	0.78 (0.76, 0.79)	0.15 (0.15, 0.16)	3,527/5,130
		RVENet+	0.87 (0.84, 0.89)	0.88 (0.85, 0.90)	0.82 (0.79, 0.85)	0.79 (0.76, 0.82)	0.16 (0.15, 0.18)	447/845
	Mean	YNHHS	0.83 (0.82, 0.83)	0.62 (0.61, 0.64)	0.73 (0.72, 0.75)	0.77 (0.76, 0.78)	0.12 (0.12, 0.12)	N/A
		RVENet+	0.81 (0.78, 0.83)	0.63 (0.61, 0.66)	0.70 (0.67, 0.73)	0.77 (0.76, 0.79)	0.15 (0.14, 0.16)	N/A
LV Diastolic Dysfunction	*Mild*	YNHHS	0.80 (0.79, 0.81)	0.63 (0.60, 0.66)	0.59 (0.57, 0.61)	0.81 (0.80, 0.83)	0.16 (0.16, 0.16)	1,214/4,071
		RVENet+	0.79 (0.76, 0.82)	0.52 (0.46, 0.58)	0.47 (0.41, 0.53)	0.87 (0.85, 0.89)	0.14 (0.13, 0.15)	182/826
	≥ Moderate	YNHHS	0.92 (0.91, 0.93)	0.52 (0.48, 0.57)	0.88 (0.85, 0.91)	0.81 (0.80, 0.82)	0.05 (0.05, 0.06)	333/4,071
		RVENet+	0.92 (0.90, 0.93)	0.67 (0.60, 0.74)	0.95 (0.92, 0.98)	0.78 (0.76, 0.81)	0.10 (0.09, 0.11)	159/826
	None	YNHHS	0.84 (0.83, 0.85)	0.90 (0.89, 0.91)	0.81 (0.79, 0.82)	0.69 (0.67, 0.71)	0.16 (0.15, 0.16)	2,524/4,071
		RVENet+	0.94 (0.92, 0.95)	0.95 (0.93, 0.96)	0.91 (0.89, 0.93)	0.81 (0.77, 0.84)	0.10 (0.09, 0.11)	485/826
	Mean	YNHHS	0.85 (0.85, 0.86)	0.68 (0.67, 0.70)	0.76 (0.75, 0.77)	0.77 (0.76, 0.78)	0.12 (0.12, 0.13)	N/A
		RVENet+	0.88 (0.87, 0.90)	0.71 (0.68, 0.74)	0.78 (0.75, 0.80)	0.82 (0.80, 0.83)	0.11 (0.11, 0.12)	N/A
Elevated LVOT PG	*≥ 20 mmHg*	YNHHS	0.94 (0.88, 0.99)	0.49 (0.33, 0.69)	0.88 (0.75, 1.00)	0.95 (0.95, 0.96)	0.00 (0.00, 0.01)	24/4,354
		RVENet+	N/A	N/A	N/A	N/A	N/A	0/944

Increased LV Size	*Mild*	YNHHS	0.83 (0.82, 0.85)	0.30 (0.26, 0.34)	0.71 (0.66, 0.75)	0.80 (0.79, 0.81)	0.06 (0.06, 0.07)	338/4,268
		RVENet+	0.78 (0.73, 0.82)	0.26 (0.20, 0.35)	0.58 (0.49, 0.67)	0.80 (0.77, 0.82)	0.07 (0.06, 0.08)	81/944
		EchoNet-LVH	0.84 (0.84, 0.85)	0.51 (0.49, 0.53)	0.56 (0.55, 0.58)	0.89 (0.88, 0.89)	0.14 (0.14, 0.15)	2,174/11,713
	≥ Moderate	YNHHS	0.94 (0.93, 0.95)	0.65 (0.60, 0.70)	0.81 (0.77, 0.85)	0.92 (0.91, 0.93)	0.04 (0.03, 0.04)	270/4,268
		RVENet+	0.91 (0.88, 0.93)	0.60 (0.52, 0.68)	0.71 (0.63, 0.78)	0.90 (0.88, 0.92)	0.06 (0.05, 0.07)	89/944
		EchoNet-LVH	0.98 (0.98, 0.99)	0.73 (0.69, 0.76)	0.95 (0.93, 0.97)	0.93 (0.92, 0.93)	0.02 (0.02, 0.02)	418/11,713
	None	YNHHS	0.91 (0.90, 0.92)	0.98 (0.98, 0.98)	0.89 (0.88, 0.90)	0.76 (0.73, 0.79)	0.08 (0.07, 0.08)	3,660/4,268
		RVENet+	0.88 (0.85, 0.90)	0.96 (0.95, 0.97)	0.88 (0.86, 0.90)	0.65 (0.59, 0.71)	0.10 (0.09, 0.11)	774/944
		EchoNet-LVH	0.89 (0.88, 0.89)	0.96 (0.96, 0.97)	0.94 (0.94, 0.94)	0.56 (0.55, 0.58)	0.13 (0.12, 0.13)	9,121/11,713
	Mean	YNHHS	0.89 (0.88, 0.90)	0.64 (0.63, 0.66)	0.80 (0.78, 0.82)	0.83 (0.82, 0.84)	0.06 (0.06, 0.06)	N/A
		RVENet+	0.85 (0.83, 0.88)	0.61 (0.58, 0.64)	0.72 (0.68, 0.76)	0.78 (0.77, 0.80)	0.08 (0.07, 0.09)	N/A
		EchoNet-LVH	0.91 (0.90, 0.91)	0.73 (0.72, 0.75)	0.82 (0.81, 0.83)	0.79 (0.79, 0.80)	0.09 (0.09, 0.10)	N/A
LV Systolic Dysfunction	*Mild*	YNHHS	0.92 (0.91, 0.93)	0.41 (0.37, 0.45)	0.89 (0.86, 0.92)	0.84 (0.83, 0.85)	0.05 (0.04, 0.05)	319/4,878
		RVENet+	0.71 (0.67, 0.75)	0.29 (0.24, 0.36)	0.50 (0.42, 0.57)	0.76 (0.74, 0.79)	0.11 (0.10, 0.13)	130/944
		EchoNet-Dynamic	0.78 (0.77, 0.79)	0.35 (0.33, 0.37)	0.58 (0.56, 0.60)	0.80 (0.79, 0.81)	0.12 (0.12, 0.12)	1,533/10,032
	≥ Moderate	YNHHS	0.98 (0.98, 0.99)	0.85 (0.82, 0.88)	0.92 (0.90, 0.95)	0.94 (0.94, 0.95)	0.02 (0.02, 0.02)	284/4,878
		RVENet+	0.99 (0.98, 0.99)	0.94 (0.91, 0.96)	0.97 (0.94, 0.99)	0.89 (0.87, 0.91)	0.03 (0.03, 0.04)	122/944
		EchoNet-Dynamic	0.94 (0.94, 0.95)	0.78 (0.76, 0.79)	0.80 (0.78, 0.82)	0.91 (0.91, 0.92)	0.06 (0.05, 0.06)	1,264/10,032
	None	YNHHS	0.97 (0.96, 0.97)	0.99 (0.99, 1.00)	0.91 (0.90, 0.92)	0.91 (0.89, 0.93)	0.04 (0.04, 0.05)	4,275/4,878
		RVENet+	0.88 (0.85, 0.90)	0.93 (0.91, 0.95)	0.85 (0.83, 0.88)	0.75 (0.70, 0.79)	0.11 (0.10, 0.12)	692/944
		EchoNet-Dynamic	0.89 (0.88, 0.90)	0.95 (0.94, 0.95)	0.90 (0.89, 0.90)	0.71 (0.69, 0.72)	0.12 (0.12, 0.13)	7,235/10,032
	Mean	YNHHS	0.96 (0.95, 0.96)	0.75 (0.73, 0.77)	0.91 (0.90, 0.92)	0.90 (0.89, 0.91)	0.04 (0.03, 0.04)	N/A
		RVENet+	0.86 (0.84, 0.88)	0.72 (0.70, 0.75)	0.77 (0.74, 0.80)	0.80 (0.78, 0.82)	0.09 (0.08, 0.09)	N/A
		EchoNet-Dynamic	0.87 (0.86, 0.88)	0.69 (0.68, 0.70)	0.76 (0.75, 0.77)	0.81 (0.80, 0.81)	0.10 (0.10, 0.10)	N/A
LV Wall Motion Abnormalities	*Any*	YNHHS	0.88 (0.86, 0.90)	0.75 (0.71, 0.78)	0.76 (0.73, 0.80)	0.85 (0.83, 0.86)	0.11 (0.10, 0.12)	348/1,541
		RVENet+	0.98 (0.97, 0.99)	0.91 (0.88, 0.94)	0.98 (0.96, 1.00)	0.71 (0.69, 0.74)	0.06 (0.06, 0.07)	116/944
Increased LV Wall Thickness	*Any*	YNHHS	0.85 (0.84, 0.86)	0.89 (0.88, 0.90)	0.82 (0.81, 0.83)	0.69 (0.67, 0.70)	0.16 (0.15, 0.16)	1,971/4,901
		RVENet+	0.83 (0.80, 0.85)	0.63 (0.57, 0.68)	0.50 (0.45, 0.56)	0.90 (0.88, 0.92)	0.14 (0.13, 0.15)	232/944
		EchoNet-LVH	0.84 (0.84, 0.85)	0.94 (0.94, 0.94)	0.84 (0.84, 0.85)	0.63 (0.62, 0.65)	0.14 (0.14, 0.14)	2,971/11,651
	≥ Moderate	YNHHS	0.91 (0.90, 0.93)	0.99 (0.99, 0.99)	0.90 (0.89, 0.90)	0.75 (0.71, 0.79)	0.04 (0.04, 0.04)	315/4,815

		RVENet+	0.91 (0.88, 0.94)	0.39 (0.30, 0.54)	0.50 (0.38, 0.62)	0.96 (0.95, 0.97)	0.04 (0.03, 0.05)	46/944
		EchoNet-LVH	0.89 (0.88, 0.90)	0.99 (0.99, 0.99)	0.92 (0.92, 0.93)	0.57 (0.54, 0.60)	0.04 (0.04, 0.05)	703/11,651
MV Regurgitation	*None-Trace*	YNHHS	0.84 (0.84, 0.85)	0.88 (0.87, 0.89)	0.85 (0.84, 0.86)	0.66 (0.64, 0.67)	0.16 (0.15, 0.16)	2,976/5,007
		RVENet+	0.91 (0.90, 0.93)	0.91 (0.88, 0.94)	0.92 (0.90, 0.94)	0.73 (0.69, 0.77)	0.12 (0.11, 0.13)	479/829
	Mild	YNHHS	0.73 (0.72, 0.75)	0.49 (0.47, 0.51)	0.70 (0.68, 0.72)	0.65 (0.63, 0.66)	0.18 (0.17, 0.18)	1,419/5,007
		RVENet+	0.78 (0.75, 0.81)	0.52 (0.46, 0.58)	0.63 (0.57, 0.68)	0.78 (0.75, 0.81)	0.16 (0.15, 0.17)	209/829
	≥ Moderate	YNHHS	0.91 (0.90, 0.91)	0.60 (0.57, 0.63)	0.74 (0.71, 0.77)	0.87 (0.86, 0.88)	0.07 (0.07, 0.08)	612/5,007
		RVENet+	0.92 (0.91, 0.94)	0.67 (0.61, 0.74)	0.88 (0.83, 0.92)	0.83 (0.80, 0.85)	0.09 (0.08, 0.09)	141/829
	Mean	YNHHS	0.83 (0.82, 0.84)	0.66 (0.64, 0.67)	0.76 (0.75, 0.77)	0.72 (0.72, 0.73)	0.14 (0.13, 0.14)	N/A
		RVENet+	0.87 (0.86, 0.89)	0.70 (0.67, 0.74)	0.81 (0.79, 0.83)	0.78 (0.76, 0.79)	0.12 (0.12, 0.13)	N/A
MV Stenosis	*Any*	YNHHS	0.96 (0.94, 0.98)	1.00 (1.00, 1.00)	0.96 (0.96, 0.97)	0.82 (0.75, 0.88)	0.01 (0.01, 0.02)	89/4,202
		RVENet+	1.00 (1.00, 1.00)	0.95 (0.85, 1.00)	1.00 (1.00, 1.00)	0.95 (0.94, 0.97)	0.01 (0.00, 0.01)	11/829
Pericardial Effusion	*Any*	YNHHS	0.91 (0.83, 0.97)	0.99 (0.98, 1.00)	0.91 (0.89, 0.93)	0.86 (0.73, 0.96)	0.02 (0.02, 0.03)	21/623
		RVENet+	0.94 (0.89, 0.98)	0.86 (0.78, 0.93)	0.68 (0.56, 0.79)	1.00 (1.00, 1.00)	0.04 (0.03, 0.05)	47/944
Elevated RA Pressure	*≥ 8 mmHg*	YNHHS	0.81 (0.80, 0.83)	0.53 (0.50, 0.56)	0.62 (0.59, 0.65)	0.82 (0.81, 0.83)	0.10 (0.10, 0.11)	722/4,646
		RVENet+	0.80 (0.78, 0.82)	0.86 (0.84, 0.88)	0.49 (0.46, 0.53)	0.96 (0.94, 0.98)	0.36 (0.34, 0.37)	531/944
Increased RA Size	*Any*	YNHHS	0.77 (0.75, 0.79)	0.88 (0.86, 0.89)	0.71 (0.69, 0.73)	0.69 (0.66, 0.72)	0.17 (0.16, 0.17)	507/1,732
		RVENet+	0.75 (0.72, 0.78)	0.50 (0.44, 0.57)	0.70 (0.65, 0.76)	0.68 (0.65, 0.71)	0.15 (0.14, 0.16)	220/944
Increased RV Size	*Mild*	YNHHS	0.78 (0.76, 0.79)	0.24 (0.22, 0.27)	0.60 (0.56, 0.63)	0.77 (0.76, 0.78)	0.08 (0.07, 0.08)	457/4,760
		RVENet+	0.64 (0.60, 0.69)	0.15 (0.12, 0.20)	0.40 (0.33, 0.49)	0.74 (0.72, 0.77)	0.09 (0.08, 0.10)	94/944
	≥ Moderate	YNHHS	0.88 (0.86, 0.90)	0.34 (0.29, 0.40)	0.72 (0.66, 0.77)	0.85 (0.84, 0.86)	0.04 (0.03, 0.04)	215/4,760
		RVENet+	0.83 (0.79, 0.86)	0.48 (0.41, 0.57)	0.64 (0.56, 0.72)	0.81 (0.79, 0.84)	0.09 (0.08, 0.11)	112/944
	None	YNHHS	0.84 (0.82, 0.85)	0.97 (0.96, 0.97)	0.83 (0.82, 0.84)	0.64 (0.62, 0.68)	0.10 (0.09, 0.10)	4,088/4,760
		RVENet+	0.78 (0.75, 0.81)	0.92 (0.91, 0.94)	0.82 (0.80, 0.85)	0.57 (0.52, 0.63)	0.15 (0.13, 0.16)	738/944
	Mean	YNHHS	0.83 (0.82, 0.84)	0.52 (0.50, 0.54)	0.72 (0.69, 0.74)	0.75 (0.74, 0.76)	0.07 (0.07, 0.07)	N/A
		RVENet+	0.75 (0.73, 0.78)	0.52 (0.49, 0.55)	0.62 (0.59, 0.66)	0.71 (0.69, 0.73)	0.11 (0.10, 0.12)	N/A
RV Systolic Dysfunction	*Any*	YNHHS	0.93 (0.91, 0.94)	0.99 (0.99, 0.99)	0.89 (0.88, 0.90)	0.80 (0.77, 0.84)	0.05 (0.04, 0.05)	371/4,743
		RVENet+	0.94 (0.92, 0.96)	0.64 (0.56, 0.73)	0.92 (0.87, 0.97)	0.82 (0.80, 0.84)	0.05 (0.04, 0.05)	76/944
TV Regurgitation	*None-Trace*	YNHHS	0.83 (0.82, 0.84)	0.85 (0.84, 0.86)	0.82 (0.80, 0.83)	0.67 (0.65, 0.68)	0.17 (0.16, 0.17)	2,886/5,076
		RVENet+	0.86 (0.84, 0.88)	0.88 (0.85, 0.90)	0.86 (0.84, 0.89)	0.71 (0.68, 0.75)	0.16 (0.15, 0.17)	532/944
	Mild	YNHHS	0.72 (0.71, 0.73)	0.50 (0.48, 0.52)	0.64 (0.62, 0.66)	0.68 (0.67, 0.69)	0.19 (0.18, 0.19)	1,580/5,076
		RVENet+	0.75 (0.72, 0.77)	0.55 (0.51, 0.61)	0.62 (0.58, 0.67)	0.76 (0.73, 0.79)	0.20 (0.19, 0.21)	310/944
	≥ Moderate	YNHHS	0.90 (0.89, 0.91)	0.61 (0.58, 0.65)	0.76 (0.73, 0.78)	0.86 (0.86, 0.87)	0.07 (0.07, 0.08)	610/5,076
		RVENet+	0.91 (0.90, 0.93)	0.58 (0.49, 0.65)	0.86 (0.81, 0.92)	0.80 (0.77, 0.82)	0.07 (0.06, 0.07)	102/944
	Mean	YNHHS	0.82 (0.81, 0.82)	0.66 (0.64, 0.67)	0.74 (0.73, 0.75)	0.74 (0.73, 0.74)	0.14 (0.14, 0.15)	N/A
		RVENet+	0.84 (0.83, 0.86)	0.67 (0.64, 0.70)	0.78 (0.76, 0.81)	0.76 (0.74, 0.77)	0.14 (0.14, 0.15)	N/A

Multi-task validation of PanEcho on diagnostic classification tasks. Results are presented on the internal YNHHS validation cohort and external RVENet+, EchoNet-Dynamic, and EchoNet-LVH cohorts. Values in parentheses represent bootstrapped 95% confidence intervals. For multi-class classification tasks, metrics are presented for each individual class, followed by the "Mean" overall performance across classes. n = number of studies meeting the label definition; N = number of non-missing labels; * = moderate or higher; ^ = severe.
AP = average precision; AUC = area under the receiver operator characteristic curve; AV = aortic valve; LA = left atrium; LV = left ventricle; LVOT = left ventricular outflow tract; PG = pressure gradient; RA = right atrium; RV = right ventricle; TV = tricuspid valve; YNHHS = Yale-New Haven Health System.

eTable 4 | Detailed Multi-task Regression Performance Evaluation

		MAE	MAD	RMSE	NMAE	r	Median (IQR)	N
Aortic Root Dimension (mm)	YNHHS	2.82 (2.77, 2.87)	2.33 (2.26, 2.39)	3.62 (3.55, 3.69)	0.08 (0.08, 0.09)	0.69 (0.68, 0.70)	3.3 (3.0-3.7)	4,968
	RVENet+	2.16 (2.08, 2.26)	1.83 (1.72, 1.94)	2.72 (2.61, 2.83)	0.07 (0.07, 0.07)	0.51 (0.47, 0.55)	3.2 (3.0-3.4)	943
AV Peak Velocity (m/s)	YNHHS	0.31 (0.30, 0.32)	0.24 (0.23, 0.24)	0.43 (0.42, 0.44)	0.19 (0.18, 0.19)	0.79 (0.78, 0.81)	1.4 (1.2-1.8)	4,499
	RVENet+	0.36 (0.34, 0.38)	0.25 (0.23, 0.26)	0.51 (0.48, 0.54)	0.23 (0.22, 0.24)	0.84 (0.81, 0.86)	1.3 (1.2-1.6)	934
LV Ejection Fraction (%)	YNHHS	4.17 (4.08, 4.25)	3.21 (3.13, 3.28)	5.55 (5.43, 5.68)	0.07 (0.07, 0.07)	0.85 (0.84, 0.86)	61.7 (57.0-65.1)	5,045
	RVENet+	4.53 (4.33, 4.71)	3.61 (3.42, 3.83)	5.83 (5.55, 6.08)	0.08 (0.08, 0.09)	0.87 (0.86, 0.89)	57.5 (52.2-61.4)	944
	EchoNet-Dynamic	5.53 (5.45, 5.62)	4.02 (3.93, 4.10)	7.70 (7.57, 7.84)	0.10 (0.10, 0.10)	0.79 (0.78, 0.80)	59.2 (51.6-64.0)	10,032
E|e' Ratio	YNHHS	1.97 (1.92, 2.01)	1.54 (1.50, 1.58)	2.62 (2.56, 2.69)	0.19 (0.19, 0.20)	0.80 (0.79, 0.81)	9.2 (7.2-12.1)	4,349
	RVENet+	2.18 (2.07, 2.31)	1.61 (1.51, 1.72)	2.95 (2.76, 3.12)	0.24 (0.23, 0.26)	0.84 (0.82, 0.86)	7.0 (5.5-10.6)	728
Global Longitudinal Strain (%)	YNHHS	1.89 (1.78, 2.01)	1.56 (1.39, 1.67)	2.44 (2.28, 2.60)	0.10 (0.09, 0.11)	0.65 (0.59, 0.71)	19.0 (18.0-21.0)*	461
	RVENet+	2.65 (2.54, 2.75)	2.28 (2.16, 2.40)	3.28 (3.15, 3.41)	0.16 (0.15, 0.17)	0.82 (0.80, 0.84)	17.6 (13.3-20.7)	866
IVSd (mm)	YNHHS	1.30 (1.27, 1.33)	1.02 (1.00, 1.05)	1.79 (1.71, 1.90)	0.13 (0.12, 0.13)	0.70 (0.68, 0.72)	1.0 (0.9-1.1)	5,045
	RVENet+	1.58 (1.52, 1.65)	1.28 (1.19, 1.36)	2.02 (1.94, 2.09)	0.15 (0.14, 0.15)	0.73 (0.70, 0.75)	1.1 (0.9-1.2)	944
	EchoNet-LVH	1.34 (1.32, 1.36)	1.06 (1.04, 1.07)	1.79 (1.76, 1.82)	0.13 (0.13, 0.13)	0.69 (0.68, 0.70)	1.0 (0.9-1.2)	11,681
LAIDs (mm)	YNHHS	3.98 (3.90, 4.05)	3.24 (3.15, 3.33)	5.09 (5.00, 5.19)	0.10 (0.10, 0.10)	0.79 (0.78, 0.80)	4.0 (3.5-4.5)	4,832
	RVENet+	4.03 (3.81, 4.25)	3.29 (3.00, 3.54)	5.17 (4.88, 5.44)	0.10 (0.09, 0.10)	0.79 (0.77, 0.81)	4.0 (3.4-4.6)	601
LA Volume (cm^3)	YNHHS	9.41 (9.22, 9.60)	7.89 (7.63, 8.11)	11.86 (11.63, 12.10)	0.16 (0.16, 0.17)	0.78 (0.77, 0.79)	55.2 (43.0-70.0)	4,107
	RVENet+	13.39 (12.88, 13.89)	12.10 (11.11, 13.11)	16.25 (15.63, 16.87)	0.20 (0.20, 0.21)	0.80 (0.78, 0.82)	64.5 (46.2-91.3)	845
LV ED Volume (cm^3)	YNHHS	18.97 (18.48, 19.46)	14.26 (13.79, 14.83)	25.97 (25.19, 26.73)	0.17 (0.16, 0.17)	0.86 (0.85, 0.87)	105.1 (80.6-136.9)	3,449
	RVENet+	34.83 (33.12, 36.56)	26.96 (25.30, 28.59)	45.48 (43.30, 47.68)	0.25 (0.24, 0.26)	0.81 (0.78, 0.83)	128.9 (104.0-166.1)	944
	EchoNet-Dynamic	25.36 (24.96, 25.79)	17.65 (17.28, 18.00)	36.32 (35.57, 37.07)	0.28 (0.28, 0.28)	0.70 (0.69, 0.71)	82.1 (62.2-108.3)	10,032
LV ES Volume (cm^3)	YNHHS	10.41 (10.06, 10.76)	6.94 (6.67, 7.18)	16.06 (15.39, 16.79)	0.21 (0.21, 0.22)	0.90 (0.90, 0.91)	40.1 (28.9-56.4)	3,435
	RVENet+	19.18 (18.09, 20.32)	13.07 (12.26, 14.12)	28.11 (26.23, 30.03)	0.28 (0.28, 0.29)	0.90 (0.88, 0.91)	55.0 (41.9-78.0)	944
	EchoNet-Dynamic	14.31 (14.01, 14.62)	8.40 (8.21, 8.61)	23.42 (22.73, 24.09)	0.33 (0.33, 0.34)	0.82 (0.81, 0.83)	33.6 (23.7-49.1)	10,032
LVIDd (mm)	YNHHS	3.78 (3.70, 3.85)	3.10 (3.02, 3.16)	4.82 (4.73, 4.90)	0.08 (0.08, 0.08)	0.74 (0.73, 0.75)	4.6 (4.2-5.1)	5,039
	RVENet+	4.37 (4.19, 4.56)	3.45 (3.25, 3.67)	5.60 (5.36, 5.85)	0.09 (0.08, 0.09)	0.80 (0.78, 0.82)	4.9 (4.5-5.5)	944
	EchoNet-LVH	3.69 (3.64, 3.73)	3.02 (2.97, 3.06)	4.70 (4.65, 4.76)	0.08 (0.08, 0.08)	0.81 (0.81, 0.82)	4.6 (4.2-5.1)	11,713
LVIDs (mm)	YNHHS	3.38 (3.32, 3.45)	2.67 (2.60, 2.74)	4.42 (4.33, 4.52)	0.11 (0.11, 0.11)	0.80 (0.78, 0.81)	3.0 (2.6-3.4)	5,028
	RVENet+	4.40 (4.20, 4.62)	3.43 (3.21, 3.75)	5.65 (5.42, 5.90)	0.13 (0.12, 0.13)	0.86 (0.84, 0.87)	3.3 (2.8-3.8)	814
	EchoNet-LVH	3.45 (3.40, 3.50)	2.71 (2.66, 2.77)	4.52 (4.46, 4.59)	0.11 (0.10, 0.11)	0.86 (0.85, 0.86)	3.1 (2.7-3.6)	11,268
LVOT Diameter	YNHHS	1.35 (1.32, 1.38)	1.07 (1.03, 1.10)	1.76 (1.72, 1.80)	0.07 (0.06, 0.07)	0.59 (0.58, 0.61)	2.0 (1.9-2.2)	3,719

(mm)	RVENet+	1.54 (1.47, 1.61)	1.34 (1.25, 1.44)	1.92 (1.84, 2.00)	0.08 (0.07, 0.08)	0.53 (0.47, 0.58)	2.0 (1.9-2.2)	614
LVPWd (mm)	YNHHS	1.15 (1.13, 1.17)	0.93 (0.91, 0.96)	1.48 (1.45, 1.50)	0.11 (0.11, 0.12)	0.68 (0.66, 0.69)	1.0 (0.9-1.1)	5,040
	RVENet+	1.05 (1.01, 1.09)	0.90 (0.84, 0.94)	1.33 (1.28, 1.39)	0.11 (0.11, 0.12)	0.68 (0.65, 0.71)	0.9 (0.8-1.0)	944
	EchoNet-LVH	1.28 (1.26, 1.29)	1.06 (1.04, 1.08)	1.63 (1.61, 1.64)	0.13 (0.13, 0.13)	0.64 (0.63, 0.65)	1.0 (0.9-1.1)	11,672
LV Stroke Volume (cm^3)	YNHHS	11.70 (11.43, 11.98)	9.21 (8.89, 9.41)	15.31 (14.94, 15.69)	0.18 (0.18, 0.18)	0.77 (0.76, 0.78)	62.0 (48.3-79.0)	3,395
	RVENet+	17.33 (16.53, 18.11)	13.73 (12.72, 14.58)	22.70 (21.65, 23.70)	0.24 (0.23, 0.25)	0.62 (0.58, 0.66)	69.5 (56.4-84.6)	944
RA Transverse Dimension (mm)	YNHHS	4.67 (4.57, 4.78)	3.94 (3.80, 4.07)	5.94 (5.82, 6.08)	0.12 (0.12, 0.13)	0.62 (0.60, 0.64)	3.7 (3.3-4.2)	2,897
	RVENet+	5.15 (4.95, 5.35)	4.43 (4.14, 4.77)	6.35 (6.12, 6.57)	0.13 (0.12, 0.13)	0.57 (0.53, 0.61)	4.0 (3.5-4.5)	944
RVIDd (mm)	YNHHS	3.97 (3.87, 4.06)	3.25 (3.13, 3.36)	5.09 (4.98, 5.20)	0.13 (0.12, 0.13)	0.57 (0.54, 0.59)	3.1 (2.7-3.5)	3,430
	RVENet+	3.85 (3.67, 4.03)	3.05 (2.83, 3.27)	5.03 (4.80, 5.25)	0.12 (0.12, 0.13)	0.58 (0.53, 0.62)	3.1 (2.8-3.5)	944
RV Systolic Pressure (mmHg)	YNHHS	5.98 (5.84, 6.14)	4.60 (4.45, 4.73)	8.11 (7.88, 8.35)	0.20 (0.20, 0.21)	0.68 (0.66, 0.70)	27.5 (22.5-35.0)	3,298
	RVENet+	6.72 (6.30, 7.12)	5.03 (4.65, 5.43)	9.34 (8.67, 10.00)	0.21 (0.20, 0.22)	0.73 (0.70, 0.75)	28.0 (22.0-38.0)	689
RV S' (cm/s)	YNHHS	1.91 (1.86, 1.96)	1.49 (1.44, 1.54)	2.61 (2.52, 2.70)	0.15 (0.14, 0.15)	0.60 (0.58, 0.63)	12.6 (10.9-14.7)	3,631
	RVENet+	1.99 (1.90, 2.08)	1.64 (1.49, 1.76)	2.55 (2.43, 2.68)	0.16 (0.15, 0.17)	0.62 (0.59, 0.66)	13.0 (11.0-15.0)	831
TAPSE (mm)	YNHHS	3.36 (3.29, 3.43)	2.82 (2.75, 2.91)	4.28 (4.19, 4.38)	0.16 (0.15, 0.16)	0.59 (0.58, 0.61)	2.1 (1.8-2.5)	3,734
	RVENet+	3.78 (3.64, 3.92)	3.26 (3.14, 3.41)	4.68 (4.51, 4.85)	0.18 (0.17, 0.18)	0.72 (0.69, 0.74)	2.1 (1.7-2.6)	944
TV Peak Gradient (mmHg)	YNHHS	5.57 (5.44, 5.72)	4.26 (4.09, 4.41)	7.55 (7.34, 7.78)	0.22 (0.21, 0.22)	0.64 (0.62, 0.66)	24.0 (19.0-30.0)	3,515
	RVENet+	6.48 (6.08, 6.85)	5.10 (4.63, 5.41)	8.86 (8.26, 9.42)	0.26 (0.25, 0.27)	0.68 (0.65, 0.71)	22.0 (17.0-30.0)	689

Multi-task international validation of PanEcho on continuous parameter estimation tasks. Results are presented on the internal YNHHS validation cohort and external RVENet+, EchoNet-Dynamic, and EchoNet-LVH cohorts. Values in parentheses represent bootstrapped 95% confidence intervals unless otherwise specified. Median and IQR are computed from non-missing ground truth measurements in each cohort. * = Global longitudinal strain values are presented as positive to aid readability; MAD = median absolute deviation; MAE = mean absolute error; N = number of non-missing measurements; NMAE = normalized mean absolute error; r = Pearson's correlation coefficient.

AV = aortic valve; ED = end-diastolic; ES = end-systolic; IVSd = interventricular septum thickness at diastole; IQR = interquartile range; LA = left atrium; LV = left ventricle; LAIDs = left atrial internal diameter at systole; LVIDd = left ventricular internal diameter at diastole; LVIDs = left ventricular internal diameter at systole; LVOT = left ventricular outflow tract; LVPWd = left ventricular posterior wall thickness at diastole; RA = right atrium; RV = right ventricle; RVIDd = right ventricular internal diameter at diastole; RV S' = right ventricular systolic excursion velocity; TV = tricuspid valve; YNHHS = Yale-New Haven Health System.

eTable 5 | Detailed Performance Evaluation at the Point of Care

		Dataset	AUC	AP	Sensitivity	Specificity	Brier Score	n/N
AV Stenosis	*None*	Abbreviated TTE	0.91 (0.90, 0.93)	0.99 (0.98, 0.99)	0.90 (0.89, 0.91)	0.77 (0.73, 0.80)	0.06 (0.06, 0.07)	2,907/3,285
		POCUS	0.87 (0.85, 0.89)	0.79 (0.77, 0.80)	0.75 (0.70, 0.80)	0.97 (0.96, 0.98)	0.87 (0.86, 0.88)	2,109/2,311
	Mild-Moderate	Abbreviated TTE	0.89 (0.88, 0.91)	0.49 (0.44, 0.54)	0.76 (0.72, 0.80)	0.87 (0.86, 0.88)	0.06 (0.06, 0.07)	309/3,285
		POCUS	0.85 (0.83, 0.88)	0.76 (0.70, 0.81)	0.77 (0.76, 0.79)	0.20 (0.18, 0.23)	0.32 (0.29, 0.36)	164/2,311
	Severe	Abbreviated TTE	0.96 (0.94, 0.97)	0.42 (0.33, 0.53)	0.83 (0.75, 0.90)	0.94 (0.93, 0.95)	0.02 (0.01, 0.02)	69/3,285
		POCUS	0.86 (0.81, 0.92)	0.55 (0.41, 0.68)	0.96 (0.95, 0.96)	0.17 (0.12, 0.23)	0.26 (0.19, 0.34)	38/2,311
	Mean	Abbreviated TTE	0.92 (0.91, 0.93)	0.63 (0.60, 0.67)	0.83 (0.80, 0.86)	0.86 (0.85, 0.87)	0.05 (0.04, 0.05)	N/A
		POCUS	0.86 (0.84, 0.88)	0.70 (0.65, 0.75)	0.83 (0.81, 0.84)	0.45 (0.43, 0.47)	0.48 (0.46, 0.51)	N/A
AV Structure	*Any*	Abbreviated TTE	0.77 (0.70, 0.84)	1.00 (0.99, 1.00)	0.75 (0.74, 0.76)	0.68 (0.56, 0.79)	0.01 (0.01, 0.01)	37/3,219
		POCUS	0.70 (0.56, 0.85)	0.95 (0.94, 0.95)	0.30 (0.00, 0.56)	1.00 (1.00, 1.00)	0.97 (0.97, 0.97)	10/2,567
Increased LA Size	*Mild*	Abbreviated TTE	0.74 (0.72, 0.76)	0.36 (0.33, 0.39)	0.68 (0.65, 0.71)	0.67 (0.66, 0.68)	0.12 (0.11, 0.13)	570/3,608
		POCUS	0.63 (0.61, 0.65)	0.50 (0.46, 0.55)	0.67 (0.65, 0.68)	0.17 (0.16, 0.19)	0.26 (0.23, 0.28)	403/3,310
	≥ Moderate	Abbreviated TTE	0.87 (0.86, 0.88)	0.56 (0.53, 0.60)	0.74 (0.72, 0.77)	0.82 (0.81, 0.83)	0.10 (0.09, 0.10)	598/3,608
		POCUS	0.78 (0.77, 0.80)	0.67 (0.64, 0.69)	0.74 (0.72, 0.75)	0.42 (0.40, 0.45)	0.52 (0.49, 0.54)	741/3,310
	None	Abbreviated TTE	0.82 (0.81, 0.83)	0.91 (0.90, 0.91)	0.72 (0.71, 0.74)	0.76 (0.74, 0.78)	0.16 (0.15, 0.16)	2,440/3,608
		POCUS	0.72 (0.71, 0.74)	0.62 (0.60, 0.64)	0.72 (0.70, 0.74)	0.81 (0.79, 0.82)	0.70 (0.69, 0.72)	2,166/3,310
	Mean	Abbreviated TTE	0.81 (0.80, 0.82)	0.61 (0.59, 0.63)	0.72 (0.70, 0.73)	0.75 (0.74, 0.76)	0.13 (0.12, 0.13)	N/A
		POCUS	0.71 (0.70, 0.73)	0.60 (0.58, 0.61)	0.71 (0.70, 0.72)	0.47 (0.46, 0.48)	0.49 (0.48, 0.50)	N/A
LV Diastolic Dysfunction	*Mild*	Abbreviated TTE	0.79 (0.78, 0.80)	0.63 (0.61, 0.66)	0.61 (0.58, 0.64)	0.79 (0.77, 0.80)	0.17 (0.16, 0.17)	892/2,825
		POCUS	0.65 (0.63, 0.68)	0.73 (0.69, 0.76)	0.48 (0.45, 0.50)	0.34 (0.31, 0.36)	0.46 (0.43, 0.49)	473/1,769
	≥ Moderate	Abbreviated TTE	0.91 (0.89, 0.92)	0.51 (0.46, 0.57)	0.85 (0.81, 0.88)	0.83 (0.82, 0.84)	0.06 (0.05, 0.06)	247/2,825
		POCUS	0.83 (0.82, 0.85)	0.79 (0.75, 0.82)	0.74 (0.72, 0.76)	0.48 (0.45, 0.51)	0.60 (0.57, 0.62)	416/1,769
	None	Abbreviated TTE	0.84 (0.83, 0.85)	0.89 (0.88, 0.90)	0.80 (0.78, 0.81)	0.70 (0.68, 0.72)	0.16 (0.16, 0.17)	1,686/2,825
		POCUS	0.80 (0.79, 0.82)	0.59 (0.56, 0.62)	0.85 (0.83, 0.87)	0.79 (0.77, 0.82)	0.68 (0.65, 0.70)	880/1,769
	Mean	Abbreviated	0.85 (0.84, 0.86)	0.68 (0.66, 0.70)	0.75 (0.73, 0.77)	0.77 (0.76, 0.78)	0.13 (0.12, 0.13)	N/A

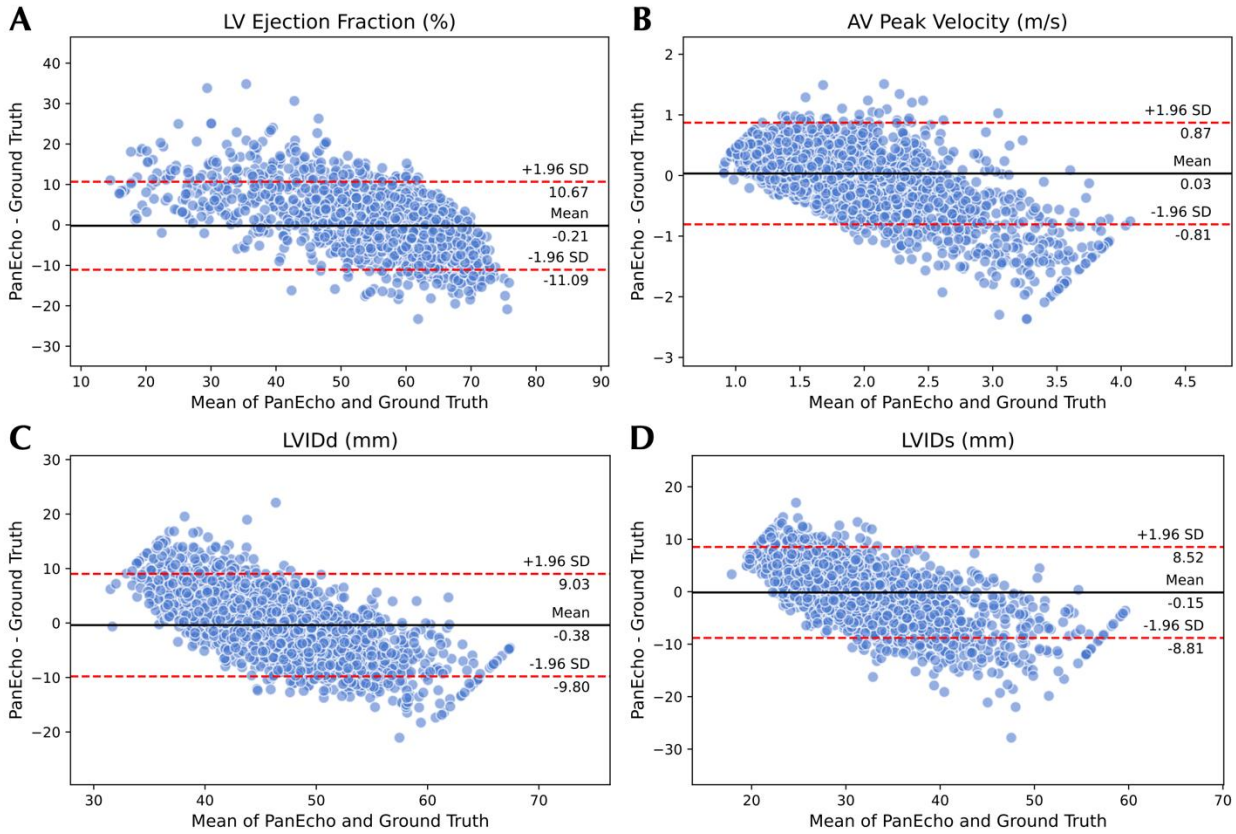
		TTE						
		POCUS	0.76 (0.75, 0.78)	0.70 (0.68, 0.72)	0.69 (0.68, 0.70)	0.54 (0.52, 0.55)	0.58 (0.56, 0.59)	N/A
Elevated LVOT PG	*≥ 20 mmHg*	Abbreviated TTE	0.94 (0.87, 0.99)	0.54 (0.37, 0.71)	0.73 (0.56, 0.88)	0.96 (0.95, 0.96)	0.01 (0.00, 0.01)	22/3,096
		POCUS	N/A	N/A	N/A	N/A	N/A	2/2,433
Increased LV Size	*Mild*	Abbreviated TTE	0.83 (0.81, 0.85)	0.30 (0.26, 0.35)	0.69 (0.64, 0.74)	0.80 (0.79, 0.81)	0.07 (0.06, 0.07)	253/2,981
		POCUS	0.64 (0.62, 0.67)	0.50 (0.46, 0.55)	0.68 (0.67, 0.70)	0.20 (0.17, 0.22)	0.28 (0.26, 0.31)	351/2,632
	≥ Moderate	Abbreviated TTE	0.94 (0.92, 0.95)	0.63 (0.58, 0.69)	0.76 (0.71, 0.81)	0.92 (0.91, 0.92)	0.04 (0.03, 0.04)	201/2,981
		POCUS	0.89 (0.87, 0.90)	0.72 (0.69, 0.75)	0.89 (0.88, 0.90)	0.64 (0.61, 0.67)	0.68 (0.65, 0.70)	547/2,632
	None	Abbreviated TTE	0.90 (0.89, 0.91)	0.98 (0.97, 0.98)	0.88 (0.87, 0.89)	0.72 (0.69, 0.76)	0.08 (0.07, 0.09)	2,527/2,981
		POCUS	0.85 (0.84, 0.86)	0.88 (0.86, 0.89)	0.67 (0.65, 0.70)	0.84 (0.82, 0.85)	0.86 (0.85, 0.87)	1,734/2,632
	Mean	Abbreviated TTE	0.89 (0.88, 0.90)	0.64 (0.62, 0.66)	0.78 (0.75, 0.80)	0.81 (0.80, 0.82)	0.06 (0.06, 0.07)	N/A
		POCUS	0.79 (0.78, 0.81)	0.70 (0.68, 0.72)	0.75 (0.74, 0.76)	0.56 (0.55, 0.57)	0.61 (0.59, 0.62)	N/A
LV Systolic Dysfunction	*Mild*	Abbreviated TTE	0.91 (0.89, 0.92)	0.36 (0.32, 0.42)	0.83 (0.79, 0.87)	0.85 (0.84, 0.86)	0.05 (0.04, 0.05)	223/3,440
		POCUS	0.70 (0.68, 0.72)	0.71 (0.68, 0.75)	0.58 (0.57, 0.60)	0.21 (0.20, 0.23)	0.33 (0.31, 0.35)	430/3,114
	≥ Moderate	Abbreviated TTE	0.98 (0.97, 0.98)	0.81 (0.77, 0.85)	0.86 (0.82, 0.90)	0.95 (0.95, 0.96)	0.02 (0.02, 0.03)	212/3,440
		POCUS	0.93 (0.92, 0.94)	0.89 (0.87, 0.91)	0.80 (0.79, 0.82)	0.59 (0.57, 0.62)	0.71 (0.69, 0.73)	755/3,114
	None	Abbreviated TTE	0.96 (0.95, 0.97)	0.99 (0.99, 0.99)	0.91 (0.90, 0.92)	0.86 (0.83, 0.89)	0.05 (0.04, 0.05)	3,005/3,440
		POCUS	0.89 (0.88, 0.90)	0.71 (0.69, 0.73)	0.87 (0.85, 0.88)	0.90 (0.88, 0.91)	0.79 (0.78, 0.80)	1,929/3,114
	Mean	Abbreviated TTE	0.95 (0.94, 0.95)	0.72 (0.70, 0.74)	0.87 (0.85, 0.88)	0.89 (0.88, 0.90)	0.04 (0.04, 0.04)	N/A
		POCUS	0.84 (0.83, 0.85)	0.77 (0.76, 0.79)	0.75 (0.74, 0.76)	0.57 (0.56, 0.58)	0.61 (0.60, 0.62)	N/A
LV Wall Motion Abnormalities	*Any*	Abbreviated TTE	0.86 (0.84, 0.88)	0.72 (0.67, 0.76)	0.73 (0.68, 0.78)	0.83 (0.80, 0.85)	0.11 (0.10, 0.12)	249/1,096
		POCUS	0.77 (0.75, 0.79)	0.81 (0.78, 0.83)	0.53 (0.50, 0.56)	0.58 (0.55, 0.61)	0.68 (0.65, 0.70)	630/1,415
Increased LV Wall Thickness	*Any*	Abbreviated TTE	0.85 (0.84, 0.86)	0.89 (0.88, 0.90)	0.82 (0.80, 0.83)	0.70 (0.68, 0.72)	0.15 (0.15, 0.16)	1,465/3,434
		POCUS	0.75 (0.73, 0.76)	0.37 (0.35, 0.39)	0.90 (0.89, 0.92)	0.83 (0.80, 0.85)	0.51 (0.49, 0.53)	1,206/2,735
	≥ Moderate	Abbreviated TTE	0.93 (0.91, 0.94)	0.99 (0.99, 0.99)	0.90 (0.89, 0.91)	0.79 (0.74, 0.83)	0.04 (0.04, 0.05)	250/3,355
		POCUS	0.85 (0.83, 0.87)	0.70 (0.68, 0.71)	0.83 (0.79, 0.86)	0.97 (0.97, 0.98)	0.81 (0.80, 0.82)	258/2,710
MV Stenosis	*Any*	Abbreviated	0.94 (0.92, 0.96)	1.00 (1.00, 1.00)	0.97 (0.96, 0.97)	0.69 (0.59, 0.78)	0.02 (0.01, 0.02)	67/2,915

		TTE						
		POCUS	0.92 (0.88, 0.95)	0.95 (0.95, 0.96)	0.63 (0.54, 0.72)	0.99 (0.98, 0.99)	0.97 (0.97, 0.97)	71/2,062
Pericardial Effusion	*Any*	Abbreviated TTE	0.91 (0.82, 0.97)	0.99 (0.98, 1.00)	0.91 (0.89, 0.93)	0.76 (0.60, 0.91)	0.02 (0.02, 0.03)	21/619
		POCUS	0.86 (0.84, 0.88)	0.48 (0.46, 0.50)	0.93 (0.90, 0.95)	0.98 (0.97, 0.99)	0.64 (0.63, 0.66)	305/2,796
Elevated RA Pressure	*≥ 8 mmHg*	Abbreviated TTE	0.80 (0.78, 0.82)	0.48 (0.44, 0.53)	0.62 (0.58, 0.66)	0.82 (0.81, 0.83)	0.10 (0.09, 0.11)	485/3,289
		POCUS	0.75 (0.74, 0.77)	0.96 (0.95, 0.97)	0.22 (0.20, 0.23)	0.48 (0.47, 0.50)	0.64 (0.63, 0.66)	1,200/2,775
Increased RA Size	*Any*	Abbreviated TTE	0.75 (0.72, 0.77)	0.87 (0.84, 0.89)	0.70 (0.68, 0.73)	0.70 (0.66, 0.74)	0.18 (0.17, 0.19)	374/1,267
		POCUS	0.74 (0.72, 0.77)	0.36 (0.34, 0.39)	0.89 (0.87, 0.91)	0.82 (0.79, 0.86)	0.50 (0.47, 0.53)	541/1,310
Increased RV Size	*Mild*	Abbreviated TTE	0.76 (0.74, 0.78)	0.24 (0.21, 0.27)	0.57 (0.52, 0.61)	0.77 (0.76, 0.79)	0.09 (0.08, 0.10)	359/3,360
		POCUS	0.67 (0.64, 0.69)	0.79 (0.76, 0.82)	0.46 (0.44, 0.47)	0.21 (0.19, 0.22)	0.33 (0.31, 0.35)	427/2,793
	≥ Moderate	Abbreviated TTE	0.88 (0.86, 0.90)	0.27 (0.22, 0.33)	0.69 (0.63, 0.75)	0.86 (0.85, 0.87)	0.04 (0.03, 0.04)	158/3,360
		POCUS	0.87 (0.86, 0.89)	0.97 (0.96, 0.98)	0.38 (0.37, 0.40)	0.21 (0.20, 0.23)	0.35 (0.33, 0.37)	411/2,793
	None	Abbreviated TTE	0.82 (0.80, 0.84)	0.96 (0.95, 0.96)	0.84 (0.82, 0.85)	0.61 (0.58, 0.65)	0.11 (0.10, 0.11)	2,843/3,360
		POCUS	0.82 (0.80, 0.83)	0.48 (0.46, 0.49)	0.89 (0.88, 0.91)	0.91 (0.90, 0.92)	0.63 (0.61, 0.64)	1,955/2,793
	Mean	Abbreviated TTE	0.82 (0.81, 0.83)	0.49 (0.47, 0.51)	0.70 (0.67, 0.72)	0.75 (0.74, 0.76)	0.08 (0.07, 0.08)	N/A
		POCUS	0.79 (0.77, 0.80)	0.75 (0.73, 0.76)	0.58 (0.57, 0.59)	0.44 (0.44, 0.45)	0.43 (0.42, 0.45)	N/A
RV Systolic Dysfunction	*Any*	Abbreviated TTE	0.92 (0.90, 0.93)	0.99 (0.99, 0.99)	0.90 (0.89, 0.91)	0.74 (0.69, 0.79)	0.05 (0.04, 0.05)	263/3,347
		POCUS	0.85 (0.84, 0.86)	0.71 (0.69, 0.72)	0.81 (0.79, 0.83)	0.90 (0.89, 0.91)	0.79 (0.78, 0.80)	819/2,813

Multi-task validation of PanEcho on diagnostic classification tasks. Results are presented on the internal YNHHS validation cohort and external RVENet+, EchoNet-Dynamic, and EchoNet-LVH cohorts. Values in parentheses represent bootstrapped 95% confidence intervals. For multi-class classification tasks, metrics are presented for each individual class, followed by the "Mean" overall performance across classes. n = number of studies meeting the label definition; N = number of non-missing labels; * = moderate or higher; ^ = severe.

AP = average precision; AUC = area under the receiver operator characteristic curve; AV = aortic valve; LA = left atrium; LV = left ventricle; LVOT = left ventricular outflow tract; PG = pressure gradient; RA = right atrium; RV = right ventricle; TV = tricuspid valve; YNHHS = Yale-New Haven Health System.

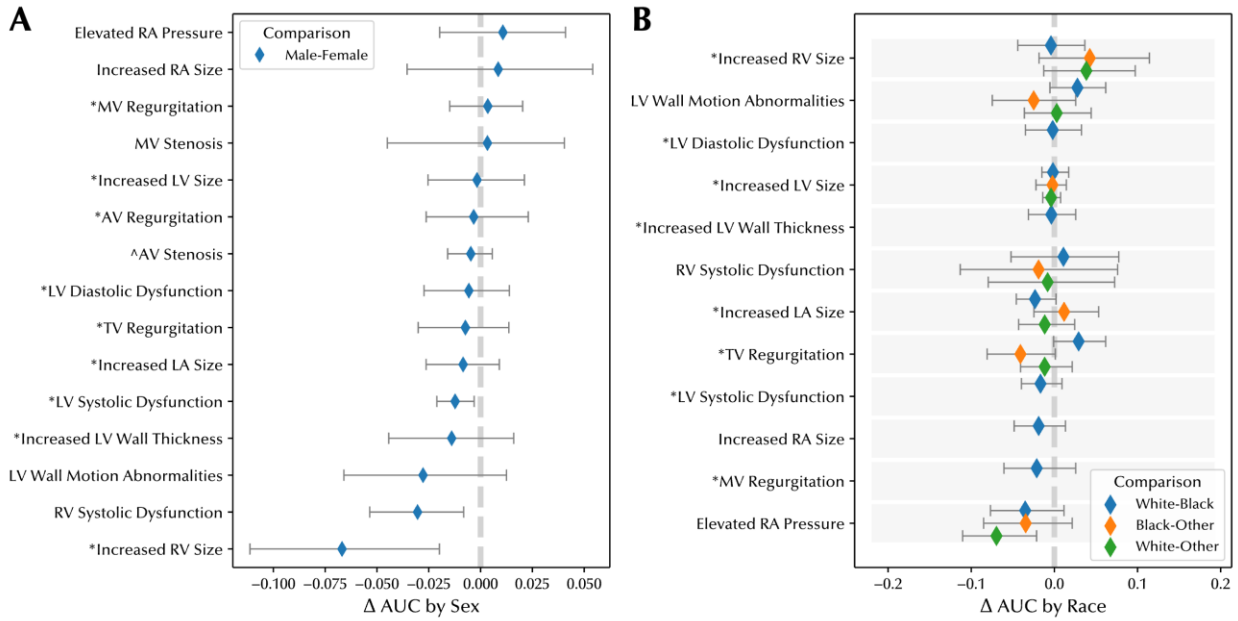
eFigure 1 | Bland-Altman Analysis for Key Echocardiographic Measurements



Bland-Altman plots for LV ejection fraction (A), AV peak velocity (B), LVIDd (C), and LVIDs (D). Analysis compares the automated PanEcho estimation vs. human expert ground truth measurement in the internal YNHHS validation set. Limits of agreement are determined by 1.96 times the standard deviation of differences.

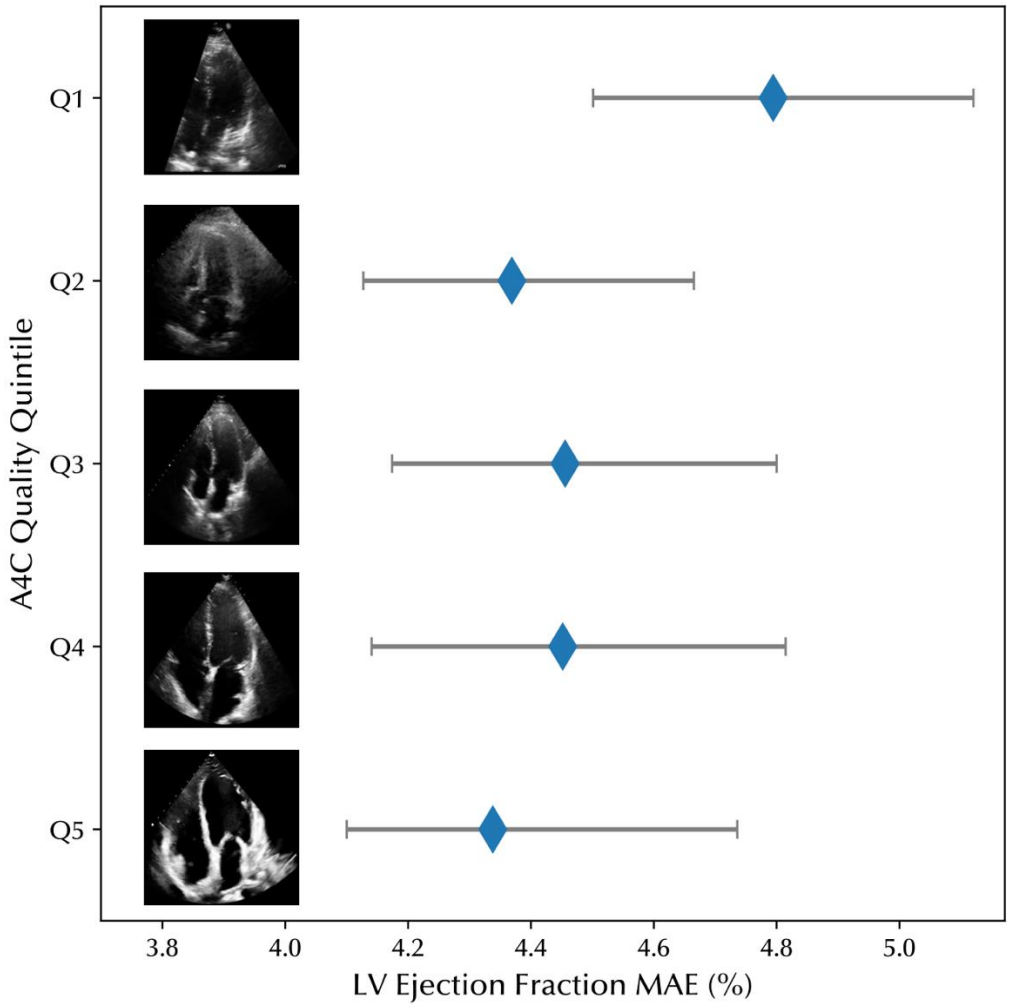
AV = aortic valve; LV = left ventricle; LVIDd = left ventricular internal diameter at diastole; LVIDs = left ventricular internal diameter at systole; YNHHS = Yale-New Haven Health System.

eFigure 2 | Group Fairness Analysis



Fairness analysis depicting the difference in AUC between different protect subgroups by sex (A) and race (B). Groupwise comparisons were excluded if the subgroup had <30 positive examples of a particular task to ensure reliable sufficient sample size. Analysis was performed in the internal YNHHS validation set. Error bars and values in parentheses represent bootstrapped 95% confidence intervals. Grey dashed line represents parity between subgroups. * = moderate or higher, ^ = severe.
AUC = area under the receiver operator characteristic curve; AV = aortic valve; LA = left atrium; LV = left ventricle; LVOT = left ventricular outflow tract; PG = pressure gradient; RA = right atrium; RV = right ventricle; TV = tricuspid valve; YNHHS = Yale-New Haven Health System.

eFigure 3 | Image Quality Analysis

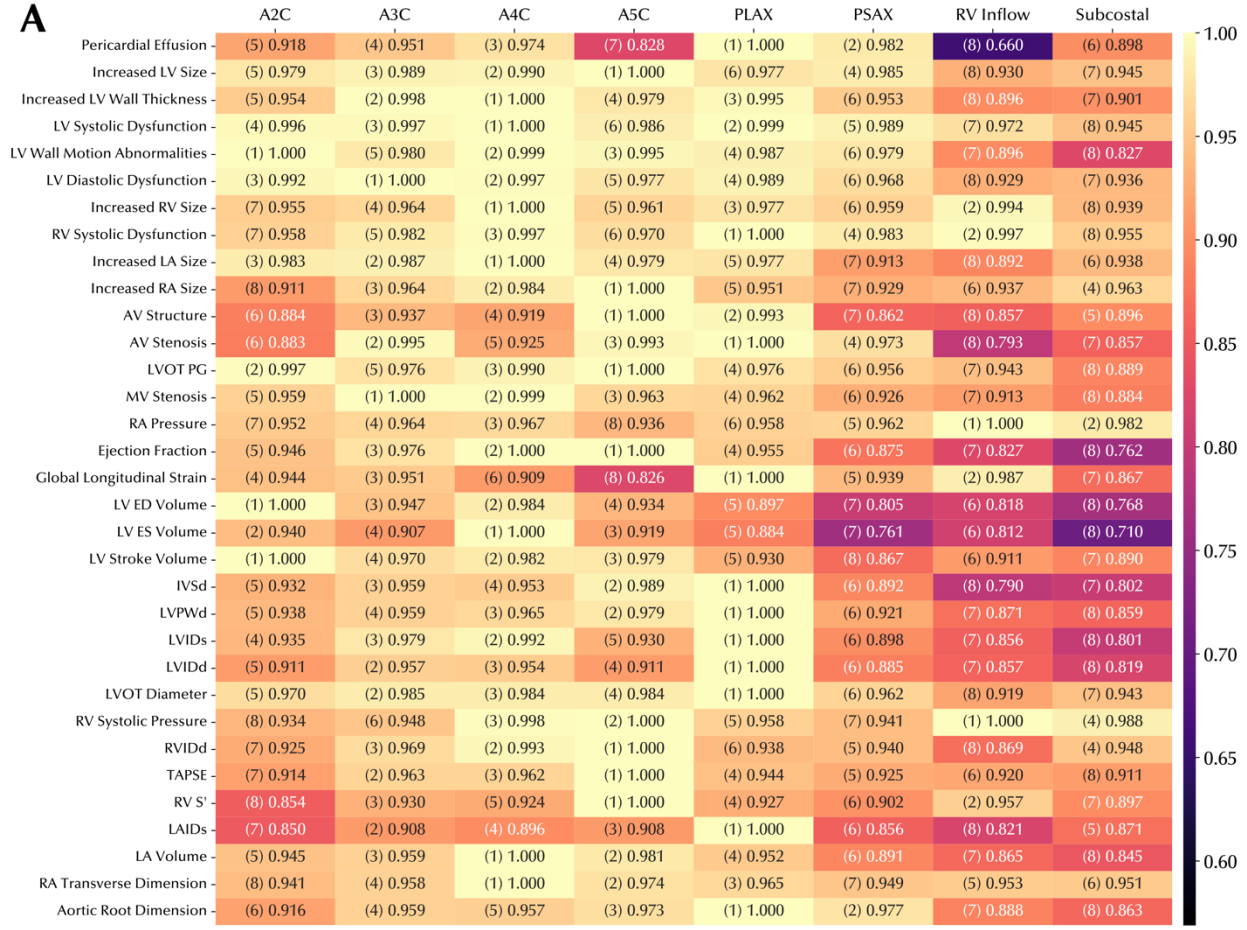


LV ejection fraction estimation performance, measured by MAE, across equal-sized "bins" of image quality. Only A4C videos from the internal YNHHS validation cohort were used for this analysis. Each bin represents a bin of image quality, defined by the quintile of A4C view confidence as predicted by the pretrained view classifier. A representative example frame from each quintile is presented alongside each row. Error bars represent bootstrapped 95% confidence intervals.
A4C = apical 4-chamber; LV = left ventricle; MAE = mean absolute error.

eFigure 4 | Detailed Task-specific View Relevance Analysis

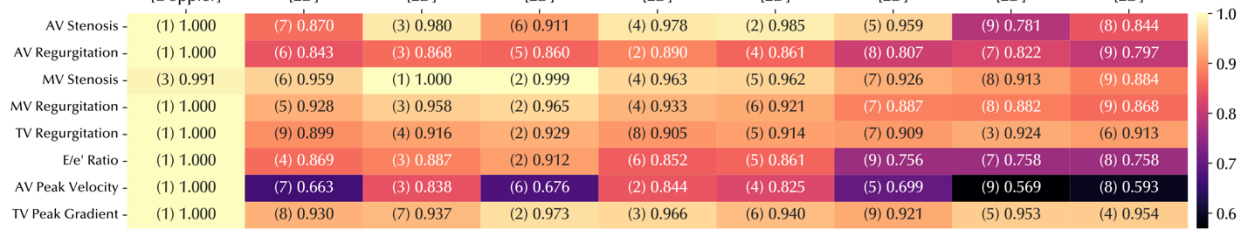
A

	A2C	A3C	A4C	A5C	PLAX	PSAX	RV Inflow	Subcostal
Pericardial Effusion	(5) 0.918	(4) 0.951	(3) 0.974	(7) 0.828	(1) 1.000	(2) 0.982	(8) 0.660	(6) 0.898
Increased LV Size	(5) 0.979	(3) 0.989	(2) 0.990	(1) 1.000	(6) 0.977	(4) 0.985	(8) 0.930	(7) 0.945
Increased LV Wall Thickness	(5) 0.954	(2) 0.998	(1) 1.000	(4) 0.979	(3) 0.995	(6) 0.953	(8) 0.896	(7) 0.901
LV Systolic Dysfunction	(4) 0.996	(3) 0.997	(1) 1.000	(6) 0.986	(2) 0.999	(5) 0.989	(7) 0.972	(8) 0.945
LV Wall Motion Abnormalities	(1) 1.000	(5) 0.980	(2) 0.999	(3) 0.995	(4) 0.987	(6) 0.979	(7) 0.896	(8) 0.827
LV Diastolic Dysfunction	(3) 0.992	(1) 1.000	(2) 0.997	(5) 0.977	(4) 0.989	(6) 0.968	(8) 0.929	(7) 0.936
Increased RV Size	(7) 0.955	(4) 0.964	(1) 1.000	(5) 0.961	(3) 0.977	(6) 0.959	(2) 0.994	(8) 0.939
RV Systolic Dysfunction	(7) 0.958	(5) 0.982	(3) 0.997	(6) 0.970	(1) 1.000	(4) 0.983	(2) 0.997	(8) 0.955
Increased LA Size	(3) 0.983	(2) 0.987	(1) 1.000	(4) 0.979	(5) 0.977	(7) 0.913	(8) 0.892	(6) 0.938
Increased RA Size	(8) 0.911	(3) 0.964	(2) 0.984	(1) 1.000	(5) 0.951	(7) 0.929	(6) 0.937	(4) 0.963
AV Structure	(6) 0.884	(3) 0.937	(4) 0.919	(1) 1.000	(2) 0.993	(7) 0.862	(8) 0.857	(5) 0.896
AV Stenosis	(6) 0.883	(2) 0.995	(5) 0.925	(3) 0.993	(1) 1.000	(4) 0.973	(8) 0.793	(7) 0.857
LVOT PG	(2) 0.997	(5) 0.976	(3) 0.990	(1) 1.000	(4) 0.976	(6) 0.956	(7) 0.943	(8) 0.889
MV Stenosis	(5) 0.959	(1) 1.000	(2) 0.999	(3) 0.963	(4) 0.962	(6) 0.926	(7) 0.913	(8) 0.884
RA Pressure	(7) 0.952	(4) 0.964	(3) 0.967	(8) 0.936	(6) 0.958	(5) 0.962	(1) 1.000	(2) 0.982
Ejection Fraction	(5) 0.946	(3) 0.976	(2) 1.000	(1) 1.000	(4) 0.955	(6) 0.875	(7) 0.827	(8) 0.762
Global Longitudinal Strain	(4) 0.944	(3) 0.951	(6) 0.909	(8) 0.826	(1) 1.000	(5) 0.939	(2) 0.987	(7) 0.867
LV ED Volume	(1) 1.000	(3) 0.947	(2) 0.984	(4) 0.934	(5) 0.897	(7) 0.805	(6) 0.818	(8) 0.768
LV ES Volume	(2) 0.940	(4) 0.907	(1) 1.000	(3) 0.919	(5) 0.884	(7) 0.761	(6) 0.812	(8) 0.710
LV Stroke Volume	(1) 1.000	(4) 0.970	(2) 0.982	(3) 0.979	(5) 0.930	(8) 0.867	(6) 0.911	(7) 0.890
IVSd	(5) 0.932	(3) 0.959	(4) 0.953	(2) 0.989	(1) 1.000	(6) 0.892	(8) 0.790	(7) 0.802
LVPWd	(5) 0.938	(4) 0.959	(3) 0.965	(2) 0.979	(1) 1.000	(6) 0.921	(7) 0.871	(8) 0.859
LVIDs	(4) 0.935	(3) 0.979	(2) 0.992	(5) 0.930	(1) 1.000	(6) 0.898	(7) 0.856	(8) 0.801
LVIDd	(5) 0.911	(2) 0.957	(3) 0.954	(4) 0.911	(1) 1.000	(6) 0.885	(7) 0.857	(8) 0.819
LVOT Diameter	(5) 0.970	(2) 0.985	(3) 0.984	(4) 0.984	(1) 1.000	(6) 0.962	(8) 0.919	(7) 0.943
RV Systolic Pressure	(8) 0.934	(6) 0.948	(3) 0.998	(2) 1.000	(5) 0.958	(7) 0.941	(1) 1.000	(4) 0.988
RVIDd	(7) 0.925	(3) 0.969	(2) 0.993	(1) 1.000	(6) 0.938	(5) 0.940	(8) 0.869	(4) 0.948
TAPSE	(7) 0.914	(2) 0.963	(3) 0.962	(1) 1.000	(4) 0.944	(5) 0.925	(6) 0.920	(8) 0.911
RV S'	(8) 0.854	(3) 0.930	(5) 0.924	(1) 1.000	(4) 0.927	(6) 0.902	(2) 0.957	(7) 0.897
LAIDs	(7) 0.850	(2) 0.908	(4) 0.896	(3) 0.908	(1) 1.000	(6) 0.856	(8) 0.821	(5) 0.871
LA Volume	(5) 0.945	(3) 0.959	(1) 1.000	(2) 0.981	(4) 0.952	(6) 0.891	(7) 0.865	(8) 0.845
RA Transverse Dimension	(8) 0.941	(4) 0.958	(1) 1.000	(2) 0.974	(3) 0.965	(7) 0.949	(5) 0.953	(6) 0.951
Aortic Root Dimension	(6) 0.916	(4) 0.959	(5) 0.957	(3) 0.973	(1) 1.000	(2) 0.977	(7) 0.888	(8) 0.863



B

	Any View [Doppler]	A2C [2D]	A3C [2D]	A4C [2D]	A5C [2D]	PLAX [2D]	PSAX [2D]	RV Inflow [2D]	Subcostal [2D]
AV Stenosis	(1) 1.000	(7) 0.870	(3) 0.980	(6) 0.911	(4) 0.978	(2) 0.985	(5) 0.959	(9) 0.781	(8) 0.844
AV Regurgitation	(1) 1.000	(6) 0.843	(3) 0.868	(5) 0.860	(2) 0.890	(4) 0.861	(8) 0.807	(7) 0.822	(9) 0.797
MV Stenosis	(3) 0.991	(6) 0.959	(1) 1.000	(2) 0.999	(4) 0.963	(5) 0.962	(7) 0.926	(8) 0.913	(9) 0.884
MV Regurgitation	(1) 1.000	(5) 0.928	(3) 0.958	(2) 0.965	(4) 0.933	(6) 0.921	(7) 0.887	(8) 0.882	(9) 0.868
TV Regurgitation	(1) 1.000	(9) 0.899	(4) 0.916	(2) 0.929	(8) 0.905	(5) 0.914	(7) 0.909	(3) 0.924	(6) 0.913
E/e' Ratio	(1) 1.000	(4) 0.869	(3) 0.887	(2) 0.912	(6) 0.852	(5) 0.861	(9) 0.756	(7) 0.758	(8) 0.758
AV Peak Velocity	(1) 1.000	(7) 0.663	(3) 0.838	(6) 0.676	(2) 0.844	(4) 0.825	(5) 0.699	(9) 0.569	(8) 0.593
TV Peak Gradient	(1) 1.000	(8) 0.930	(7) 0.937	(2) 0.973	(3) 0.966	(6) 0.940	(9) 0.921	(5) 0.953	(4) 0.954



View relevance scores depicting the relative importance of each echocardiographic view (column) for a given task (row). For each task, a score of 1 indicates the most relevant view, and values in parentheses represent the rank importance of a view. For multi-class classification tasks, only the most severe class is considered for this analysis. View relevance scores for all 39 tasks and all major 2D echocardiographic views **(A)**. View relevance scores for tasks known to be aided by Doppler imaging when including color Doppler videos with potentially unknown view **(B)**. AUC = area under the receiver operating characteristic curve; AV = aortic valve; A2C = apical 2-chamber; A3C = apical 3-chamber; A4C = apical 4-chamber; A5C = apical 5-chamber; ED = end-diastolic; ES = end-systolic; IVSd = interventricular septum thickness at diastole; LA = left atrium; LV = left ventricle; LAIDs = left atrial internal diameter at systole; LVIDd = left ventricular internal diameter at diastole; LVIDs = left ventricular internal diameter at systole; LVOT = left ventricular outflow tract; LVPWd = left ventricular posterior wall thickness at diastole; MAE = mean absolute error; PLAX = parasternal long axis; PSAX = parasternal short axis; PG = pressure gradient; RA = right atrium; RV = right ventricle; RVIDd = right ventricular internal diameter at diastole; RV S' = right ventricular systolic excursion velocity; TV = tricuspid valve.

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